

TSF

Last modified on Aug 12, 2022



The [Tenant Security Framework](#) (TSF) is Comcast’s model for micro-segmentation. TSF allows Comcast to partition resources and manage access rules in a more secure manner, and allows for greater automation and flexibility, than is available in traditional firewall solutions.

The TSF feature in the OneCloud Portal implements this model, allowing you to:

- define tenants, for logical separation of services, systems, and associated infrastructure
- create AWS and CCP networks
- model existing Comcast and TSF-MSN networks so that they can be used to provide services to tenants
- define and manage services, each in the context of AWS, CCP, Comcast legacy, or TSF-MSN, networks
- request access to other tenants’ services
- grant other tenants access to your tenant’s services

Behind the scenes, the TSF feature of the Portal is orchestrating reviews and approvals of service definitions and access policies, as well as automating infrastructure changes to enable inter-tenant communication. The Portal makes the status of review decisions and of automated infrastructure changes instantly available to you and provides a visualization tool that helps to understand the access to and from your services.

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
 - [Creating a Tenant](#)
 - [Editing a Tenant’s Setup](#)
- [Managing Networks](#)
 - [Choosing the network platform](#)
 - [Creating or importing an AWS VPC](#)
 - [Editing an AWS VPC](#)
 - [Deleting an AWS VPC](#)
 - [Creating a TSF-MSN Network](#)
 - [Activating a TSF-MSN Network](#)
 - [Editing a TSF-MSN Network](#)
 - [Downloading credentials for a TSF-MSN Network](#)
 - [Deleting a TSF-MSN Network](#)
 - [Creating a Comcast Legacy Network](#)
 - [Editing a Comcast Legacy Network](#)
 - [Deleting a Comcast](#)

AMAZON DMZ TEST TENANT

TSF Zone: DMZ

Data Classification: Public

Inbound Access

Outbound Access

My services

5 of 5 services shown

AMAZON DMZ TEST TENANT

Public rail (2 services defined)

SSS

svc1

Protected rail (3 services defined)

Amazon DMZ Test Service

SS

SSS

Private rail (2 services defined)

Amazon DMZ Test Service

Amazon DMZ Test Tenant SSH

Consuming tenants

8 of 8 tenants shown

AMAZON DMZ TEST TENANT

APORETO TEST 2

APOTEST5

EXTERNAL AUTOBAHN

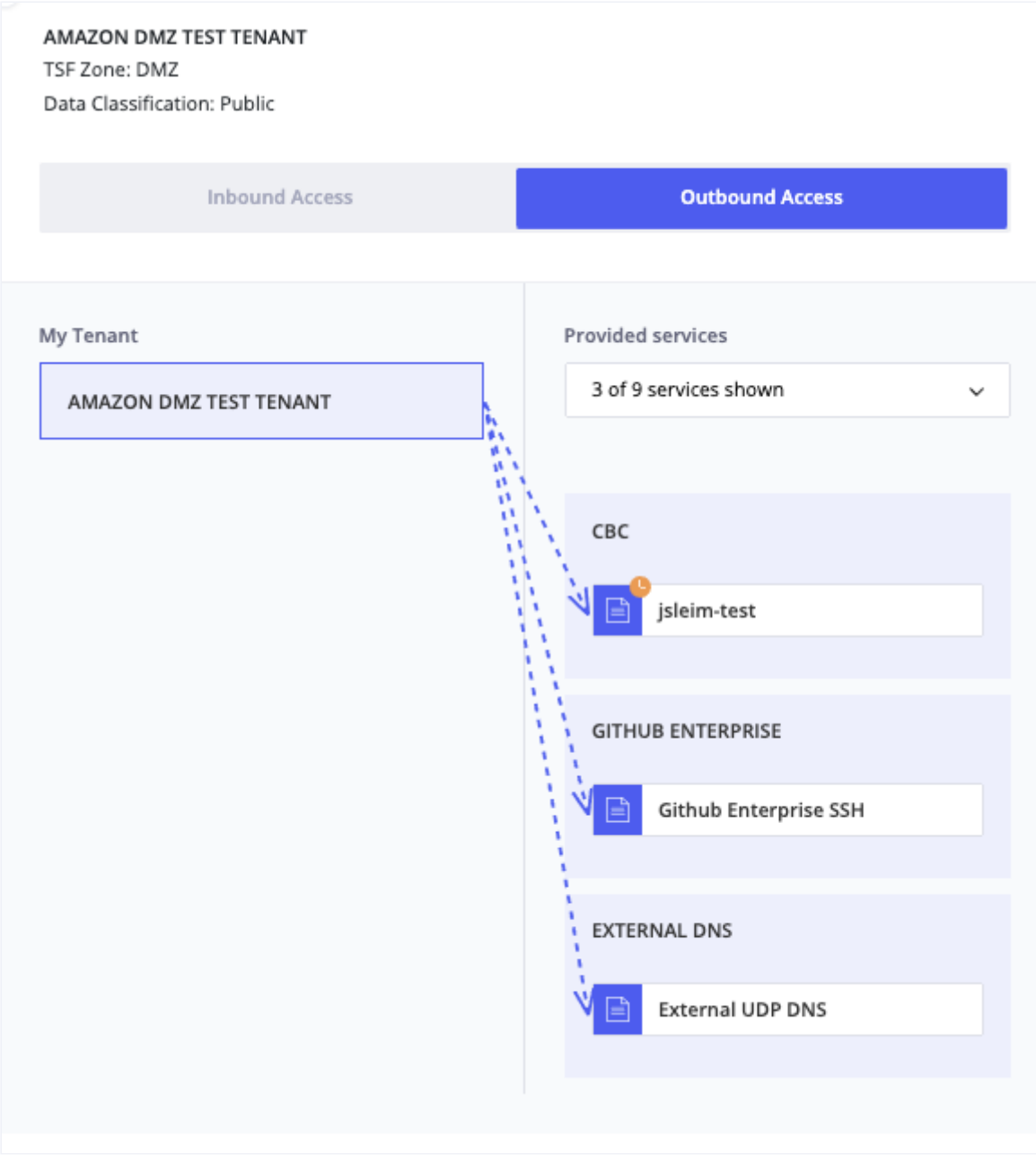
ONECLOUD

TEST AP FLOW

TEST COMCAST TENANT

TSF DMZ

VISUALIZING INBOUND ACCESS



VISUALIZING OUTBOUND ACCESS

ON THIS PAGE

[TSF Terms and Concepts](#)

[Managing Tenants](#)

[Creating a Tenant](#)

[Editing a Tenant's Setup](#)

[Managing Networks](#)

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

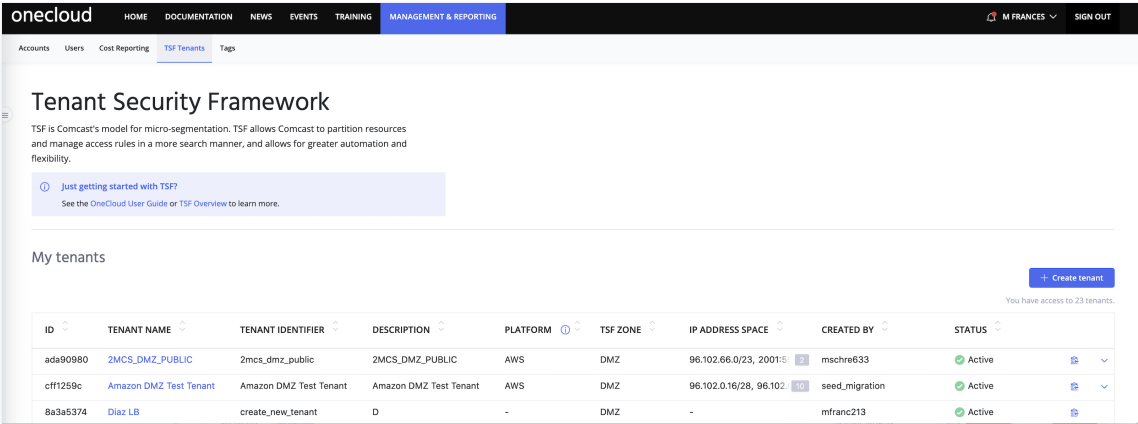
[Deleting a Comcast](#)

TSF Terms and Concepts

TERM	DEFINITION
Access Policy	A security policy that allows a TSF tenant to access a TSF service
CIDR	Classless Inter-Domain Routing - a method for defining a block of IP addresses
Rail	A logical grouping of systems within the same tenant
Service	A specification of rail(s), network port, and network protocol, by which to access an application component provided by a TSF tenant
Tenant	A logical grouping of systems and associated infrastructure within a TSF zone
TSF	Tenant Security Framework - a model for partitioning networking infrastructure. For more technical details than are provided in this user guide, see additional TSF documentation .
TSF-MSN	Tenant Security Framework Micro Segmentation Networks. For details, see our introduction to TSF-MSN .
TSF Zone	A logical grouping of systems that share similar data classification, connectivity, and security requirements. A zone defined in TSF is not the same thing as a network security zone; rather, there is a mapping between a TSF zone and one or more network security zones (see below).
VPC	Amazon's Virtual Private Cloud. For details, see Amazon's VPC documentation .

Fundamental to TSF is the concept of a tenant, which is a logical container of services, systems, and associated infrastructure within a TSF zone. Any network or service defined in TSF needs to be tied to a specific tenant. A tenant has owners who manage its resources. If you are not a tenant owner, you can view the tenant’s details but not make any changes to it.

Clicking the Portal’s **TSF Tenants** tab brings you to the TSF dashboard, where you will see tenants that you are an owner of, or that are tied to a cloud account that you are a member of.



TSF DASHBOARD

Creating a Tenant

To create a TSF tenant, click the **Create tenant** button on the TSF dashboard. The tenant create form will be displayed.

×

Create tenant

Add details about your tenant.

For more information ...

Detailed instructions for creating TSF tenants can be found [here](#).

Please note

Tenant name and owners but no other details provided in this form may be modified after tenant creation.

TENANT NAME

The name is shown to users in the OneCloud Portal. It should be concise and descriptive.

Tenant name

TENANT IDENTIFIER

The identifier is suggested for you, but edit below if needed. It may not be changed after tenant creation.

Tenant identifier

TENANT DESCRIPTION

The description is shown to users in the OneCloud Portal. Please provide a meaningful description so that others may understand what the tenant is used for.

Description

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
- [Creating a Tenant](#)
- [Editing a Tenant’s Setup](#)
- [Managing Networks](#)
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)

TSF ZONE

Which TSF Zone is best suited for the application your tenant represents? TSF Zone is based on your application's consumers, downstream dependencies, and data sensitivity.

☒ DMZ: Hosts tenants whose services are intended to accept connections from outside Comcast (e.g., the Internet).

☐ Internal: Hosts tenants whose services are intended to accept connections only from internal Comcast networks.

☐ Office: Hosts tenants whose services reside on the enterprise office network and are intended to accept connections from internal Comcast networks. Cannot be used for AWS or CCP.

☐ Sensitive: Hosts tenants whose services are subject to legal or compliance requirements, contain SNPI data, and are intended to accept connections only from internal Comcast networks.

☐ Untrusted: Hosts tenants whose services are provided by networks external to Comcast. This option is available and visible only to TSFAdmins.

DATA CLASSIFICATION

Please classify the information the tenant collects, stores, maintains, or otherwise processes. The security team uses the tenant's data classification when reviewing proposed service definitions and access policy requests. [Click here](#) for more information.

☒ Public: Information that is available to the public or that, if disclosed, would cause no harm to Comcast.

☐ Confidential: Personal information of individual customers and employees that is not otherwise Sensitive Personal Information.

☐ Sensitive Non-Public Information ("SNPI"): Information, including all Sensitive Personal Information, where the loss of confidentiality, integrity, or availability of such information could be expected to have a severe adverse effect on Comcast.

TENANT OWNERSHIP

Grant ownership to team members who will need to update settings for this tenant.

Search for individuals to add as owners.

3 tenants matched.

Search for users by name or Comcast ID ...

OneCloud

☐	NAME	COMCAST ID	EMAIL
☒	Chappel, Gary	gchapp201	Gary_Chappel@comcast.cc
☐	Couja, Salem	scouja504	Salem_Couja@comcast.cor
☐	Howard, Victor	VHowar000	Victor_Howard@cable.com
☐	Schreiber, Michael	mschre633	Michael_Schreiber2@comc

2 owners selected

Chappel, Gary

Frances, Margaret

Cancel

Create tenant

CREATING A TENANT

You will need to supply a tenant name, identifier, and description. See rules and guidance [here](#).

TSF ZONE

You will then select a TSF zone for your tenant. In order to identify the TSF zone best suited for your application, you should consider the following:

1. Who/where are your consumers?

2. Who/where are your downstream dependencies?

3. How sensitive is the data you are storing or processing?

The answers to these questions will inform the decisions you make based on the following table.

TSF ZONE	DEFINITION
DMZ	Hosts systems that are intended for exposure outside of the company. The DMZ is the only zone that can host Internet-exposed tenants.
Internal	Hosts systems that require no connectivity to external users or networks
Office	Contains users and user-owned resources. AWS VPCs can not be created in the Office zone

ON THIS PAGE

[TSF Terms and Concepts](#)

[Managing Tenants](#)

[Creating a Tenant](#)

[Editing a Tenant's Setup](#)

[Managing Networks](#)

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

[Deleting a Comcast](#)

https://onecloud.comcast.net/docs/user-guide/management-console/tsf/#tsf-terms-and-concepts

4/36

Sensitive	Hosts systems that are typically subject to legal or compliance requirements. For instance, the Payment Card Industry Data Security Standard (PCI DSS) mandates that the storage, processing, and transmission of credit card information be done in an entirely secure and segregated environment.
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DATA CLASSIFICATION

Next, you will select a data classification. The security team uses the tenant’s data classification when reviewing proposed service definitions and access policy requests. See the table below for information about the available data classifications, and see Comcast’s [VPC documentation](#) for more detailed information.

DATA CLASSIFICATION	DEFINITION
Non-Public	Data that, if compromised, corrupted, or disclosed to an unauthorized party, could have an adverse impact on the company, customers, trading partners, and/or employees but for which there is low likelihood of legal action arising from the loss of confidentiality
Public	Data that poses no risk to the company if it is made publicly available
Sensitive Non-Public	Data that, if compromised, corrupted, or disclosed to an unauthorized party, could severely harm Comcast’s reputation or business position, resulting in financial, reputational, or legal loss

The following table can help serve as a guide to TSF zone selection:

REQUIRES INBOUND INTERNET ACCESS	REQUIRES OUTBOUND INTERNET ACCESS	DATA CLASSIFICATION	TSF ZONE
Yes	Yes/No	Public/Non-Public	DMZ
Yes	Yes/No	Sensitive Non-Public	<i>Seek consultation</i>
No	Yes/No	Public/Non-Public	Internal
No	Yes	Sensitive Non-Public	<i>Seek consultation</i>
No	No	Sensitive Non-Public	Sensitive

If you are still unsure what zone is right for your tenant, don’t guess; instead, contact the [#tsf-support](#) Slack channel.

OWNERS

The last step in tenant creation is adding other owners by searching for users by name or Comcast ID. You can also search for owners of another tenant to choose from. Being a tenant owner confers the authorization to create and modify that tenant’s resources: its owners, networks, services, and access policies. A tenant must have at least one owner, but at least two are recommended. Tenant owners can be added and changed after the tenant is created (see below). *(Note that if you do not set yourself as an owner you will lose the ability to manage the tenant.)*

When these choices have been set, you can submit the request to create the tenant. The tenant will proceed through a short activation period and will be added to the list of your tenants; the dashboard will refresh automatically once the tenant is available

ON THIS PAGE

[TSF Terms and Concepts](#)

[Managing Tenants](#)

[Creating a Tenant](#)

[Editing a Tenant’s Setup](#)

[Managing Networks](#)

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

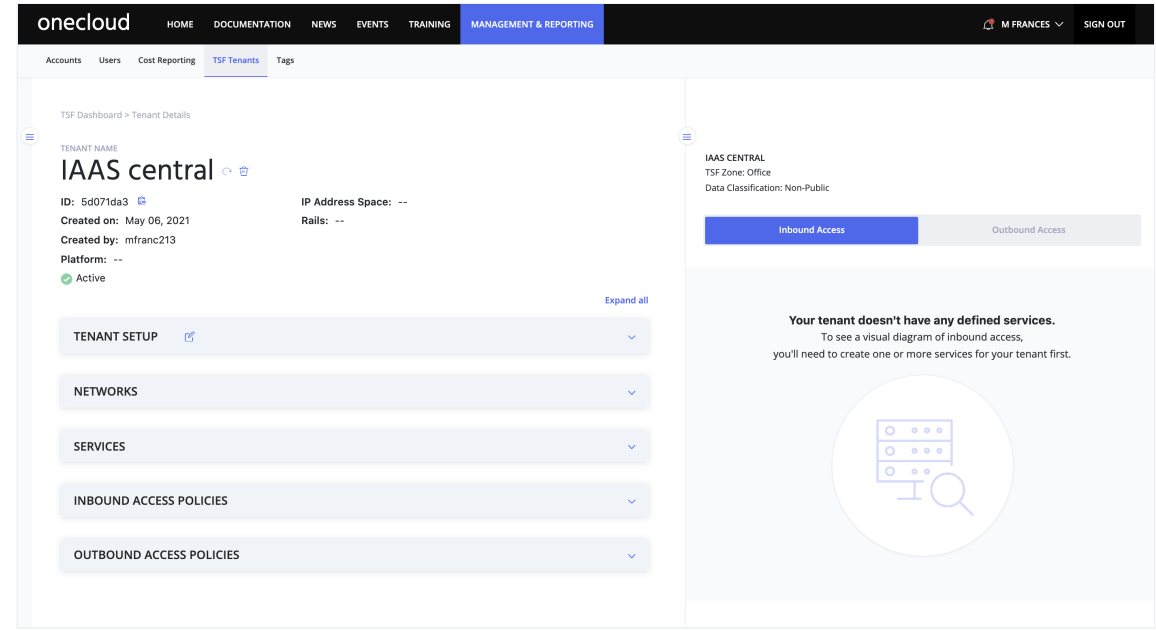
[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

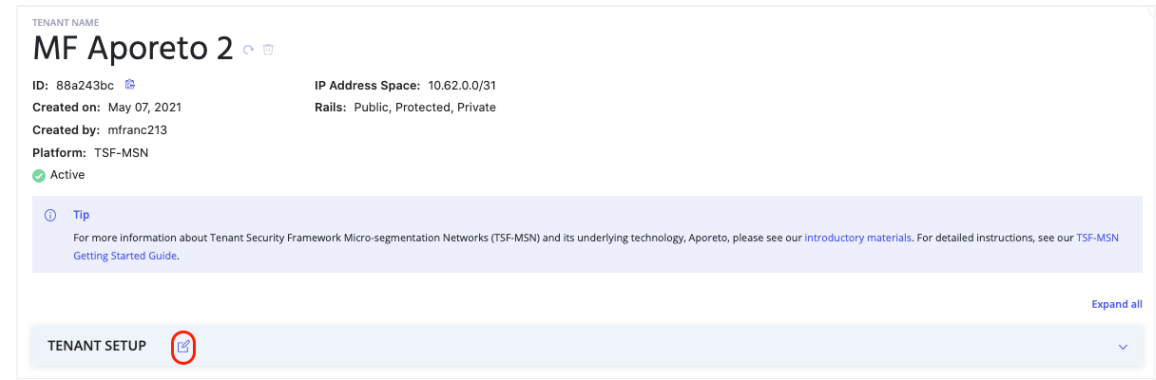
[Deleting a Comcast](#)



TENANT LANDING PAGE

Editing a Tenant’s Setup

To edit the tenant setup, click the **Edit** icon on the **Tenant Setup** accordion.



EDITING TENANT SETUP

The edit tenant setup form will be displayed.

Edit tenant setup

Edit setup for MF Aporeto 2

Tenant ID: 88a243bc

Tenant's TSF Zone: DMZ

Tenant's Data Classification: Public

TENANT NAME

The tenant name is shown to users in the OneCloud Portal. It should be concise and descriptive.

MF Aporeto 2

TENANT IDENTIFIER

The tenant identifier may not be changed. It is used to name resources by the TSF automation platforms.

mf_aporeto_2

TENANT DESCRIPTION

The description is shown to users in the OneCloud Portal. Please provide a meaningful description so that others may understand what the network is used for.

D

TENANT OWNERSHIP

Grant ownership to team members who will need to update settings for this tenant.

jslei

☐	NAME	COMCAST ID	EMAIL
☒	Sleiman, Joseph	jsleim001c	Joseph_Sleiman@comcast.net

2 owners selected

Frances, Margaret

Sleiman, Joseph

Cancel

Save changes

EDITING TENANT SETUP

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
- [Creating a Tenant](#)
- [Editing a Tenant’s Setup](#)
- [Managing Networks](#)
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)

You can change a tenant’s name and description, and you can make changes to tenant ownership by adding or removing users from the list. A tenant must have at least one owner. *(Note that if you remove yourself as owner you will lose the ability to manage the tenant.)*

The tenant identifier, TSF zone, and data classification cannot be changed.

Managing Networks

The **Networks** pane allows you to manage the tenant’s AWS, CCP, Comcast, or [TSF-MSN](#) networks. A tenant may not use more than one network platform, so once the first network has been requested, any other networks for that tenant must be of the same platform.

Things to consider when setting up a network include:

- If it’s an AWS VPC, which AWS account should be used and how do you want to structure your VPC?
- If it’s a CCP network, which CCP account should be used and how do you want to structure your network?
- If it’s a Comcast or TSF-MSN network, what are the IP addresses of the service(s) you would like to model?

Whether defining a new network or modeling an existing network, you will need to identify what rail or rails will be used for the services in your tenant. See the table below for a description of the available rails.

RAIL	DEFINITION
Public	Applications/services intended for exposure outside of your tenant are hosted on this rail. Exposing a public rail to incoming connections from the Internet will require your tenant to be in TSF’s DMZ.
Protected	Systems placed on the protected rail are intended to service requests from the public rail of your tenant.
Private	Systems placed on the private rail should not be exposed outside your tenant.

Choosing the network platform

To create the first network for your tenant, you must first specify the platform. Open the **Networks** pane and click the **Choose platform for new network** button.



CHOOSING A NETWORK PLATFORM

The platform selection form will be displayed.

ON THIS PAGE

- [TSF Terms and Concepts](#)**
- [Managing Tenants](#)**
- [Creating a Tenant](#)
- [Editing a Tenant’s Setup](#)
- [Managing Networks](#)**
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)

×

Choose a platform

For more information ...

Detailed information about the Tenant Security Framework can be found in [our cloud security documentation](#).

A tenant can use only one cloud platform type, so your selection below dictates the platform for all networks in your tenant.

What platform will your tenant use?



Comcast legacy network

You can model a legacy network in TSF if you are an owner of the tenant and the tenant is authorized to support legacy networks in TSF.



AWS

You can create an AWS VPC if you are an owner of the tenant and an application owner of at least one AWS account authorized for TSF.



CCP

You can create a CCP network if you are an owner of the tenant and an application of at least one CCP account authorized for TSF.



TSF-MSN

You can create a TSF-MSN network if you are an owner of the tenant and the tenant has been authorized for TSF.

CHOOSING A NETWORK PLATFORM

Select the desired platform. A create network form will be displayed according to the chosen platform.

Creating or importing an AWS VPC

There are two ways to model an AWS VPC in OneCloud: creating a new one or importing an existing one. For, either, the following conditions must be met:

- The tenant must be associated with the DMZ, internal, or sensitive TSF zone.
- You must be a member of at least one AWS account.

CREATING A NEW AWS VPC

If you’re creating the VPC in a new tenant, [follow the steps above](#) for choosing the platform and pick AWS.

Alternatively, if the tenant already has a network, click the **Create AWS VPC** button in the network pane.

NETWORKS

AWS VPCS

+ Create AWS VPC

Notice

Self-service allocation of DX VPCs via TSF is currently suspended. For information on how to request DX VPCs at this time, please [click here](#).

ID	VPC NAME	VPC ID	DESCRIPTI	ACCOUNT	REGION	VPC BLOC	UPDATED	STATUS	
ea6d2837	DannyTest/	vpc-33e3564d	DannyTest/	OneCloudE (6236586E	us-east-1	96.10.2	Apr 15, 2021 11:57 am	Active	

CREATING A NEW AWS VPC

The create AWS VPC form will be displayed. Select the **Create new VPC** tab.

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
- [Creating a Tenant](#)
- [Editing a Tenant’s Setup](#)
- [Managing Networks](#)
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)

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8/36

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Create AWS VPC

Add an AWS VPC to dlopez912AWSTestTenant123

Tenant's TSF Zone: DMZ

Tenant's Data Classification: Public

Create new VPCImport existing VPC

ⓘ

For more information ...

Instructions for creating an AWS VPC in the OneCloud Portal can be found in [our user guide](#).

NETWORK NAME

The network name is shown to users in the OneCloud Portal. It should be concise and descriptive.

New network

NETWORK IDENTIFIER

The identifier may not be changed. It is used to name resources by the TSF automation platforms.

new_network

AWS ACCOUNT

The account may not be changed after network creation.

Governance Test (269726743014) ▾

REGION

The network's region may not be changed after network creation.

US East (N. Virginia): us-east-1 ▾

NETWORK DESCRIPTION

The description is shown to users in the OneCloud Portal. Please provide a meaningful description so that others may understand what the network is used for.

Great network

RAILS

ⓘ

Please note

If you intend to expose services to other tenants it is recommended that you provision a public rail.

For each rail you enable, you must indicate the number of [availability zones](#) it should cover. You will also need to choose the [IPv4 subnet](#) size. The combination of these settings will determine the number of IP addresses that will be reserved for your VPC: the system will create, for each of the availability zones you have requested, an IPv4 subnet with the block size you specified as well as an IPv6 subnet of size /64.

By default you are limited to creating a network with IPv4 addresses. If you need more than IPv4 addresses, please open a support request in [OneCloud's Jira queue](#).

RAIL	ZONES	IPv6 SUBNET SIZE	IPv4 SUBNET SIZE	IPv4 COUNT
Public	1	/64	/28	16
TOTAL IPv4s				16

ADD A RAIL

RAIL

ZONES

IPv4 SUBNET SIZE

Select rail ▾

Select number of availability zones ▾

Select IPv4 subnet size ▾

Add row

Cancel

Create network

CREATING A NEW AWS VPC

You will need to supply a network name, identifier, and description. See rules and guidance [here](#).

You will then need to choose an AWS account and region. The available regions will depend on the selected account.

Now you will need to add one or more rails to the network. For each selected rail, you must specify the number of availability zones it should cover. The number of availability zones depend on the region chosen. See AWS documentation on [availability zones](#) for further information.

For each rail selected, you will also need to choose the IPv4 subnet size. This will determine the number of IPv4 addresses that will be reserved for your VPC, since a subnet will be created for each availability zone in each rail. See the AWS documentation on [VPC subnets](#) for more information. Requests for networks larger than /24 (256 IP addresses) per account/region will be sent for additional approvals.

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
- [Creating a Tenant](#)
- [Editing a Tenant's Setup](#)
- [Managing Networks](#)
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)

Click the **Create network** button to submit the request and schedule the network for activation; the VPC will generally be provisioned and available in your AWS account within one hour.

IMPORTING AN EXISTING DIRECT CONNECT VPC [🔗](#)

The portal allows end users to import Direct Connect VPCs that were created outside of OneCloud. Only [generally available Direct Connect VPCs](#) that meet the conditions described below are candidates for VPC import.

For a VPC to be imported, the following conditions must be met:

- The user requesting the import must be an Application Owner, Account Manager, or Advanced User in the VPC’s account.
- The tenant’s zone must align with the VPC’s TSF address space location.
- The VPC must have has the correct tags attached to it.
- The VPC’s subnets must be named in the correct format.
- The VPC must live in the us-east-1 and has been migrated to Transit Gateway or is in the us-west-2 region and connected to a Direct Connect Gateway.

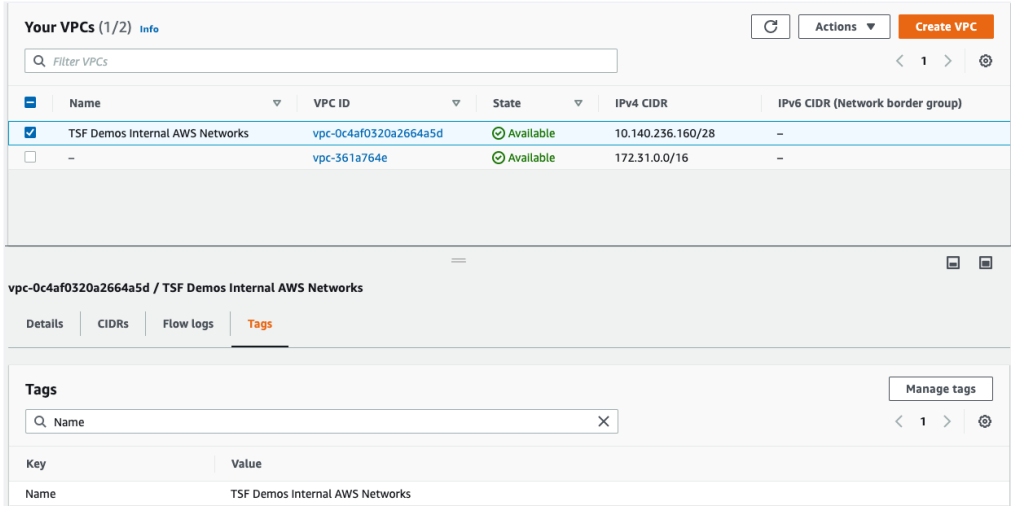
Once the VPC is imported via OneCloud, then services and access policies can be created for it via TSF Automation. Any existing access policies associated with that tenant will automatically be updated with the imported VPC’s IP space.

NOTE: The VPC cannot be edited or deleted through TSF after import. If you require modification to your VPC configuration, contact [Cloud SRE](#).

Preparing the VPC for import

Prior to requesting an import for a VPC the following steps need to be taken:

- The VPC must be tagged with the tag **Name: {VPC name}**
 - VPC name** should be set to the VPC name as you want it to appear in OneCloud, and it must include only letters, numbers, and spaces, and no special characters. **Note:** Choose wisely, as it cannot be changed later



TSF AWS IMPORT VPC - TAGGING VPC NAME

The VPC must be tagged with the tag **ComcastVPCImport: True**

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
 - [Creating a Tenant](#)
 - [Editing a Tenant’s Setup](#)
- [Managing Networks](#)
 - [Choosing the network platform](#)
 - [Creating or importing an AWS VPC](#)
 - [Editing an AWS VPC](#)
 - [Deleting an AWS VPC](#)
 - [Creating a TSF-MSN Network](#)
 - [Activating a TSF-MSN Network](#)
 - [Editing a TSF-MSN Network](#)
 - [Downloading credentials for a TSF-MSN Network](#)
 - [Deleting a TSF-MSN Network](#)
 - [Creating a Comcast Legacy Network](#)
 - [Editing a Comcast Legacy Network](#)
 - [Deleting a Comcast](#)

Your VPCs (1/2) [Info](#)

↻
Actions ▾
Create VPC

< 1 > ⚙️

	Name ▾	VPC ID ▾	State ▾	IPv4 CIDR	IPv6 CIDR (Netw
☒	TSF Demos Internal AWS Networks	vpc-0c4af0320a2664a5d	✔ Available	10.140.236.160/28	-
☐	-	vpc-361a764e	✔ Available	172.31.0.0/16	-

vpc-0c4af0320a2664a5d / TSF Demos Internal AWS Networks

Details
CIDRs
Flow logs
Tags

Tags

✕

Key	Value
ComcastVPCImport	True

ON THIS PAGE

[**TSF Terms and Concepts**](#)

[**Managing Tenants**](#)

[Creating a Tenant](#)

[Editing a Tenant's Setup](#)

[**Managing Networks**](#)

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

TSF AWS IMPORT VPC - TAGGING VPC

Subnets must be named in the same format as those created via OneCloud

Portal: Name: {VPC Name}-{rail}-{value}, where

Any spaces in the VPC name should *not* be removed.

Rail is set to **public**, **protected**, or **private**

Value must be unique for a given rail and should be a zero-based incrementing number per rail (eg. **{VPC name}-public-0**, **{VPC Name}-public-1**, **{VPC Name}-private-0**, etc.).

Subnets (1/5) Info					Refresh Actions	
Filter subnets						
☐	Name	Subnet ID	State	VPC		
☒	TSF Demos Internal AWS Networks-public-0	subnet-0965627a9ab0cccd	Available	vpc-0c4af0320a2664a5d TSF Demos Internal AWS Networks	Creating a Comcast Legacy Network Editing a Comcast Legacy Network Deleting a Comcast Legacy Network	

TSF AWS IMPORT VPC - NAMING SUBNETS

Importing the VPC

If you're importing the VPC into a new tenant, [follow the steps above](#) for choosing the platform, and pick AWS.

Alternatively, if the tenant already has a network, click the **Create AWS VPC** button in the network pane.

NETWORKS									
AWS VPCS									
+ Create AWS VPC									
Notice Self-service allocation of DX VPCs via TSF is currently suspended. For information on how to request DX VPCs at this time, please click here.									
ID	VPC NAME	VPC ID	DESCRIPTION	ACCOUNT	REGION	VPC BLOCK	UPDATED	STATUS	
ea6d2837	DannyTest/	vpc-33e3564d	DannyTest/	OneCloudE (6236586E	us-east-1	96.10.1.2	Apr 15, 2021 11:57 am	Active	

IMPORTING AN AWS VPC

The import AWS VPC modal will be displayed; choose the **Import existing VPC** tab.

ON THIS PAGE

TSF Terms and Concepts

Managing Tenants

Creating a Tenant

Editing a Tenant's Setup

Managing Networks

Choosing the network platform

Creating or importing an AWS VPC

Editing an AWS VPC

Deleting an AWS VPC

Creating a TSF-MSN Network

Activating a TSF-MSN Network

Editing a TSF-MSN Network

Downloading credentials for a TSF-MSN Network

Deleting a TSF-MSN Network

Creating a Comcast Legacy Network

Editing a Comcast Legacy Network

Deleting a Comcast

×

Create AWS VPC

Add an AWS VPC to dlopez912AWSTestTenant123

Tenant's TSF Zone: DMZ

Tenant's Data Classification: Public

Create new VPC

Import existing VPC

For more information ...

Instructions for importing an AWS VPC can be found in [our user guide](#).

Please note

An imported VPC cannot be edited or deleted through the OneCloud Portal. If you require modification to your VPC configuration, contact [our Public Cloud SRE team](#).

AWS ACCOUNT

Choose the account to which the VPC will be assigned.

Governance Test (269726743014) ▾

REGION

Choose the VPC's current region.

US East (N. Virginia): us-east-1 ▾

VPC

Governance Test (vpc-081b9a8f66efb22b9) ▾

ADDRESS SPACE

96.103.140.192/27, 2001:558:fd01:3c00::/56

VPC ID

vpc-081b9a8f66efb22b9

SUBNETS

RAIL	NAME	ZONE	IPV4 CIDR	IPV6 CIDR	IPV4 COUN
Public	public-0	us-east-1a	96.103.140.192/28	2001:558:fd01:3c00::/64	16
Public	public-1	us-east-1b	96.103.140.208/28	2001:558:fd01:3c01::/64	16
TOTAL IPV4s					32

AFFECTED SERVICES AND INBOUND ACCESS POLICIES

The import of this network will not affect any of the tenant's services.

AFFECTED OUTBOUND ACCESS POLICIES

PROVIDING TENANT	SERVICE	RAILS	PROTOCOL	PORTS	ICMP TYPE
dlopez912ComcastTenant	dlopez912Con	Protected	TCP	22	

Cancel

Import network

IMPORTING AN AWS VPC

You will first need to select the AWS account and the region the VPC lives in. The Portal will fetch the list of importable VPCs for that account from that region. Once these are loaded, the VPC list will be available for selection.

After you have selected the desired VPC, the VPC details will be displayed along with the tenant’s existing services and access policies, if any, that would be affected by the import.

Once you have confirmed the settings and affected resources, click the **Import network** button to start the import process.

Should you have any questions or run in to any issues, please reach out in the [#tsf-support](#) Slack channel.

Editing an AWS VPC [🔗](#)

Provided an AWS VPC is in the active state and was not imported, its name and description may be edited, and CIDRs may be added to it. *Note: Existing CIDRs cannot be modified or removed. If you need to make changes to existing CIDRs of your VPC, please*

ON THIS PAGE

[TSF Terms and Concepts](#)

[Managing Tenants](#)

[Creating a Tenant](#)

[Editing a Tenant’s Setup](#)

[Managing Networks](#)

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

[Deleting a Comcast](#)

<https://onecloud.comcast.net/docs/user-guide/management-console/tsf/#tsf-terms-and-concepts>

12/36

To start, click the **edit** icon next to the desired VPC.

NETWORKS

AWS VPCS

+ Create AWS VPC

+ Import AWS VPC

Notice

Self-service allocation of DX VPCs via TSF is currently suspended. For information on how to request DX VPCs at this time, please [click here](#).

Please note

Resources that were deleted more than 1 day ago or failed more than 7 days ago are not displayed. [Click to show](#).

ID	VPC NAME	VPC ID	DESCRIPTION	ACCOUNT	REGION	VPC BLOCKS	UPDATED	STATUS	
b9daefc	network 2	vpc-23337cfa	Desc	Governance Test (26972674301)	us-east-1	96.102.64	May 05, 2021 12:02 pm	Active	
49559c21	Network testing refresh	vpc-8c42e6e4	D	Governance Test (26972674301)	us-east-1	96.102.64	May 02, 2021 9:53 am	Active	

EDITING AN AWS VPC

The edit form will be displayed. Make the desired changes.

Edit AWS VPC

Edit OneCloud ElasticSearch's network Elasticsearch VPC

Network ID: 1c5c5792

Tenant's TSF Zone: Internal

Rails: Public, Protected

Tenant's Data Classification: Non-Public

VPC ID: vpc-01bacb58b8cfc8c9

VPC Blocks: 10.140.159.0/27, 10.140.191.80/28

For more information ...

Instructions for updating AWS VPCs in the OneCloud Portal can be found in [our user guide](#).

NETWORK NAME

The network name is shown to users in the OneCloud Portal. It should be concise and descriptive.

Elasticsearch VPC

NETWORK IDENTIFIER

The identifier may not be changed. It is used to name resources by the TSF automation platforms.

es_vpc_prod

AWS ACCOUNT

The AWS account may not be changed after network creation.

OneCloudEngineering (623658651045)

REGION

The network region may not be changed after network creation.

us-west-2

NETWORK DESCRIPTION

The description is shown to users in the OneCloud Portal. Please provide a meaningful description so that others may understand what the network is used for.

Very good network

SUBNETS

For each CIDR in the network, you will need to choose the [subnet](#) size. This will determine the number of IP addresses that will be reserved for your VPC.

By default you are limited to creating a network with 256 IP addresses. If you need more than 256 IP addresses, please open a support request in [OneCloud's Jira queue](#).

NAME	RAIL	ZONE	V4 CIDR	V6 CIDR	IPV4 COU	STATUS
public-0	Public	us-east-1a	96.102.64.1	2001:558:fc:/64	16	Active
public-1	Public	us-east-1a	/27	/64	32	
48						

ADD A CIDR

RAIL

ZONE

V4 CIDR

Select rail

Select zone

Select block size

Add row

Cancel

Save changes

EDITING AN AWS VPC

Once saved, changes to the CIDRs of an AWS VPC will trigger updates to all associated access policies.

ON THIS PAGE

[TSF Terms and Concepts](#)

[Managing Tenants](#)

[Creating a Tenant](#)

[Editing a Tenant's Setup](#)

[Managing Networks](#)

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

[Deleting a Comcast](#)

Provided an AWS VPC is in the active state and was not imported, may be deleted. *Note: If there are existing access policies or services that are used by other networks or if the deleted network is the last one in the tenant you will not be able to delete a network*

To start, click the **delete** icon next to the desired VPC.

NETWORKS

AWS VPCS

+ Create AWS VPC

+ Import AWS VPC

Notice

Self-service allocation of DX VPCs via TSF is currently suspended. For information on how to request DX VPCs at this time, please [click here](#).

Please note

Resources that were deleted more than 1 day ago or failed more than 7 days ago are not displayed. [Click to show](#).

ID	VPC NAME	VPC ID	DESCRIPTION	ACCOUNT	REGION	VPC BLOCKS	UPDATED	STATUS		
b9daefc	network 2	vpc-23337cfa	Desc	Governance Test (26972674301)	us-east-1	96.102.64	May 05, 2021 12:02 pm	Active		
49559c21	Network testing refresh	vpc-8c42e6e4	D	Governance Test (26972674301)	us-east-1	96.102.64	May 02, 2021 9:53 am	Active		

DELETING AN AWS VPC

A confirmation screen will be displayed. The services and access policies that would be affected by the deletion will be shown for review.

Confirm AWS VPC deletion

Network ID: b9daefc

Rails: Public

VPC ID: vpc-23337cfa

VPC Blocks: 96.102.64.16/28, 2001:558:fd01::/56

Tenant's TSF Zone: DMZ

Tenant's Data Classification: Public

AFFECTED SERVICES AND INBOUND ACCESS POLICIES

SERVICE	RAILS	PROTOCOL	PORTS	ICMP TYPE	CONSUMING TENANTS
Test network delete 5	Public	TCP	1234		MF AWS no IPv6 2, 5

AFFECTED OUTBOUND ACCESS POLICIES

PROVIDING TENANT	SERVICE	RAILS	PROTOCOL	PORTS	ICMP TYPE
Diaz LB	Service 2	Public	TCP	12334	
MF AWS no IPv6 2	Test network delete 5	Public	TCP	1234	
MF AWS with IPv6 8	Service 2	Public	UDP	444	

No, I changed my mind

Delete network

DELETING AN AWS VPC

Click the **Delete network** to start the deletion process. Once an AWS VPC is deleted, the associated access policies will be updated.

Creating a TSF-MSN Network

Note
A TSF-MSN network is a mechanism to record the IP addresses and/or address blocks that will be used for TSF-MSN. While TSF-MSN is an identity-based solution, IP is needed to ensure communication with systems that do not support identity.

A tenant may have at most one TSF-MSN network. In a new tenant, [follow the steps above](#) for choosing the platform, and pick TSF-MSN. The create AWS TSF-MSN network form will be displayed.

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
- [Creating a Tenant](#)
- [Editing a Tenant's Setup](#)
- [Managing Networks](#)
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)

×

Create TSF-MSN network

Add a TSF-MSN network to MF new tenant

Tenant's TSF Zone: DMZ

Tenant's Data Classification: Public

ⓘ

For more information ...

Instructions for creating a TSF-MSN network in the OneCloud Portal can be found in [our user guide](#).

NETWORK NAME

The network name is shown to users in the OneCloud Portal. It should be concise and descriptive.

Network name

NETWORK IDENTIFIER

The identifier may not be changed. It is used to name resources by the TSF automation platforms.

Network ID

NETWORK DESCRIPTION

The description is shown to users in the OneCloud Portal. Please provide a meaningful description so that others may understand what the network is used for.

Description

CIDRNETS

SUBNETS

NAME	RAIL	CIDR BLOCK	IP COUNT
------	------	------------	----------

ADD A CIDR

NAME

RAIL

CIDR BLOCK

Enter CIDR name

Select rail

Enter CIDR block

Add row

Cancel

Create network

CREATING A TSF-MSN NETWORK

You will need to supply a network name, identifier, and description. See rules and guidance [here](#).

You will then need to add one or more CIDRs. A CIDR may represent a single host (i.e., /32) or an address block, and it must adhere to the following rules:

1. All CIDRs in the network must fall within the same network security zone. To determine which network security zone your CIDR is in, consult the list of infrastructure blocks defined by the Network Engineering Local Internet Registry Group. The table on that page maps infrastructure blocks to services; see below for the mapping of each service to a network security zone.

NETWORK SECURITY ZONE	SERVICES
Agile	Agile
Internal	Internal Services, IWAP, UET Infrastructure, VOD Infrastructure, Anycast - Internal Services
Office	Enterprise, Enterprise - NBCU, Anycast - Enterprise
Public	External Services, External Services - Voice CO, Infrastructure Delivery, LoopBack, Loopback CCS, P2P, Anycast - CPE, Anycast - External Services
Sensitive	Backend Firewall, Anycast - Backend Firewall

ON THIS PAGE

[TSF Terms and Concepts](#)

[Managing Tenants](#)

[Creating a Tenant](#)

[Editing a Tenant's Setup](#)

[Managing Networks](#)

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

[Deleting a Comcast](#)

https://onecloud.comcast.net/docs/user-guide/management-console/tsf/#tsf-terms-and-concepts

15/36

ULA	ULA
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2. The network security zone must align with the tenant’s [TSF zone](#) per the following table.

TSF ZONE	NETWORK SECURITY ZONES
DMZ	Agile, Internet, Public (Blue)
Internal	Internal, UEN, ULA (Green)
Office	Office (Red)
Sensitive	Sensitive (Black)

Click the **Create Network** button to schedule the network for provisioning in the TSF-MSN control plane.

NETWORKS

TSF-MSN NETWORKS

+ Create TSF-MSN network

NAME	DESCRIPTION	LAST UPDATED	STATUS	ID	
Network 1	Fabulous Network	Jun 28, 2021 11:27 am	Pending Security Approval	Please contact the Security Change Review Group if not approved within two business days.	2b22b7e2

CREATING A TSF-MSN NETWORK

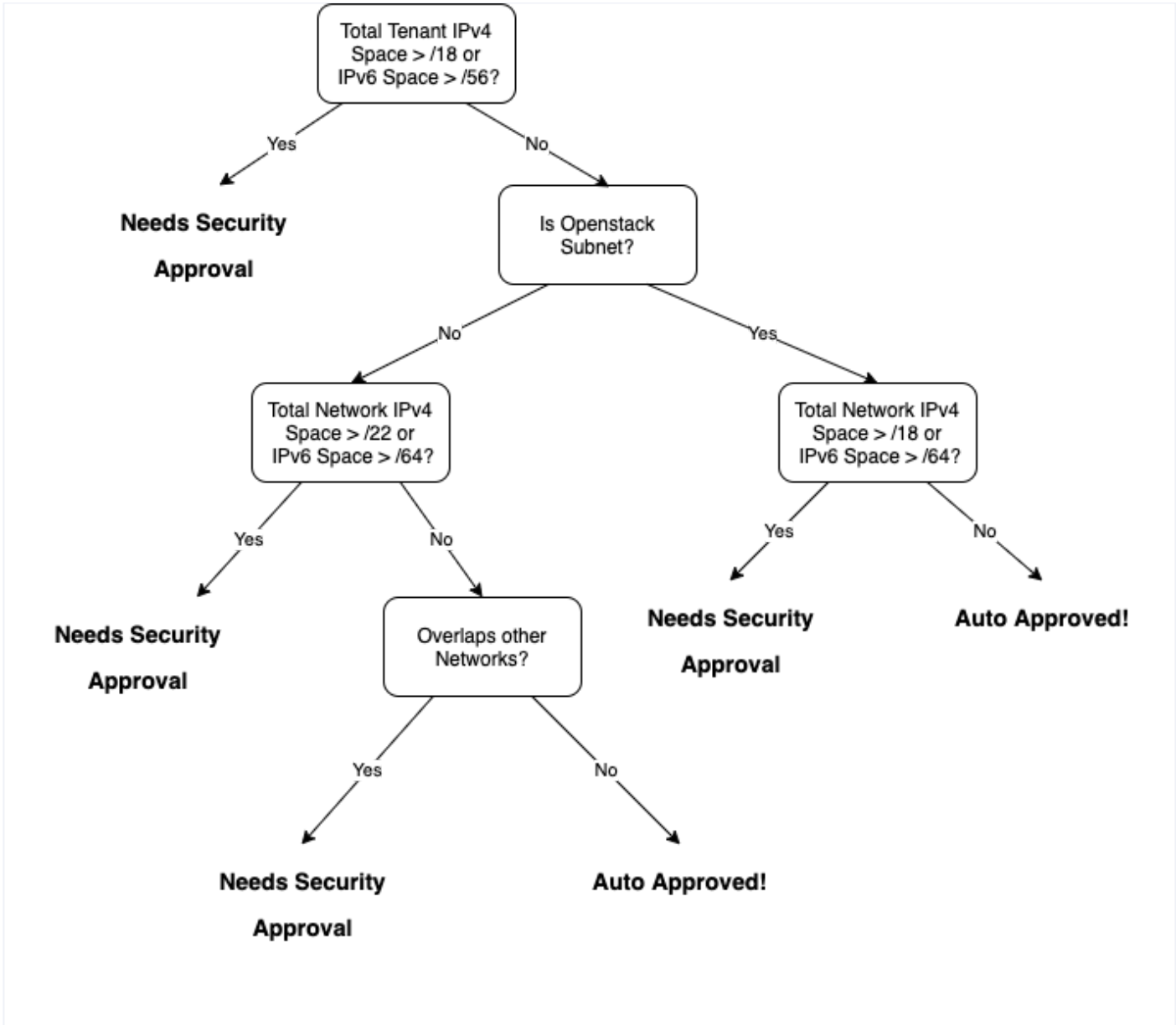
TSF-MSN networks go through multiple states before activation, as follows:

NETWORK STATUS	DEFINITION
Pending Security Approval	Network needs to be reviewed by security (rules specified in the graph below)
Pending pre-activation	Network is not yet provisioned.
Pre-active	Network is provisioned but in a log-only mode. The TSF-MSN agent will not block TCP or UDP traffic while in this state. You will need to [manually activate the network]#activating-a-tsf-msn-network (see next section).
Active	Network is fully provisioned and in enforcement mode. Any traffic that is not defined with an access policy is blocked.

AUTO-APPROVAL AND SECURITY REVIEW REQUIRED CASES

ON THIS PAGE

- [TSF Terms and Concepts](#)**
- [Managing Tenants](#)**
- [Creating a Tenant](#)
- [Editing a Tenant’s Setup](#)
- [Managing Networks](#)**
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)



APORETO NETWORK APPROVAL FLOW

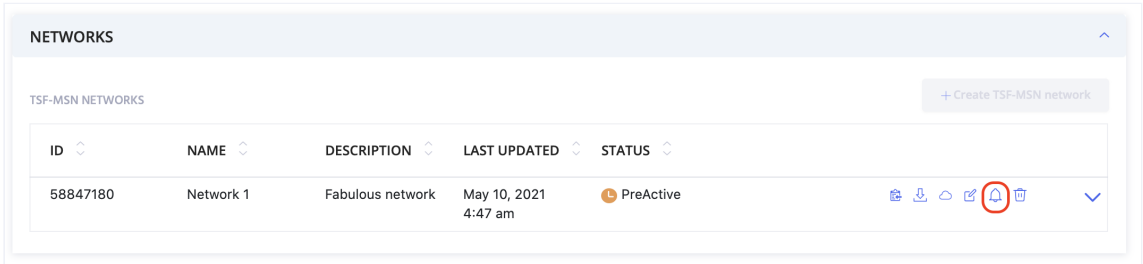
Activating a TSF-MSN Network

Warning

Any network access that is not defined in the tenant’s access policies will lose connectivity upon activation.

When a TSF-MSN network is first created in OneCloud, it enters a pre-active state, during which it is governed by a “discovery-mode” policy, which tells the agent to allow all communication and log the traffic. Finalizing the provisioning of a TSF-MSN network in this state requires an activation step, which removes the discovery-mode policy from the network. At this point, network traffic without a defined policy is blocked.

To start, click the **activate** icon next to the network.



ACTIVATING A TSF-MSN NETWORK

A confirmation screen will be displayed.

ON THIS PAGE

[TSF Terms and Concepts](#)

[Managing Tenants](#)

[Creating a Tenant](#)

[Editing a Tenant’s Setup](#)

[Managing Networks](#)

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

[Deleting a Comcast](#)

Confirm network activation

Activate MF Aporeto 2's network Network 1

Network ID: d382ca45

Tenant's TSF Zone: DMZ

Rails:

Tenant's Data Classification: Public

Warning

Any access to or from this network that is not defined in this tenant's access policies will be removed when the network is activated. Also, note that activation will enable agent monitoring. If the agent becomes unresponsive, the SIRT may eject the host, which would block its communications.

Please review the following, all of which will be enabled upon network activation.

NETWORK

ID	NAME	LAST UPDATED	STATUS
d382ca45	Network 1	May 07, 2021 12:01 pm	PreActive

SERVICES

ID	NAME	LAST UPDATED	STATUS
53ba7703	SSH	May 10, 2021 4:19 am	PreActive

INBOUND ACCESS POLICIES

This network has no inbound access policies.

OUTBOUND ACCESS POLICIES

ID	NAME	LAST UPDATED	STATUS
98c167ff	mf_aporeto_2 - service_2	May 10, 2021 4:23 am	PreActive

Cancel

Activate network

ACTIVATING A TSF-MSN NETWORK

The confirmation screen displays the existing networks, services, and inbound and outbound access policies that will be affected by the importing of the network. Click the **Activate network** button, at which point the network will enter the pending activation state. Activation itself takes about 5 minutes.

Editing a TSF-MSN Network

Provided a TSF-MSN is in the active state, its name and description may be edited, and CIDRs may be added or removed. Based on the modification, security review may be required as specified in [this graph](#).

To start, click the **edit** icon.

NETWORKS

TSF-MSN NETWORKS

+ Create TSF-MSN network

ID	NAME	DESCRIPTION	LAST UPDATED	STATUS
d382ca45	Network 1	D	May 10, 2021 4:24 am	Active

EDITING A TSF-MSN NETWORK

The edit TSF-MSN form will be displayed.

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
- [Creating a Tenant](#)
- [Editing a Tenant's Setup](#)
- [Managing Networks](#)
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)

×

Edit TSF-MSN network

Edit MF TSF-MSN tenant's network Network 1

Network ID: 58847180

Tenant's TSF Zone: DMZ

Rails: --

Tenant's Data Classification: Public

For more information ...

Instructions for updating TSF-MSN networks in the OneCloud Portal can be found in [our user guide](#).

NETWORK NAME

The network name is shown to users in the OneCloud Portal. It should be concise and descriptive.

Network 1

NETWORK IDENTIFIER

The identifier may not be changed. It is used to name resources by the TSF automation platforms.

network_1

NETWORK DESCRIPTION

The description is shown to users in the OneCloud Portal. Please provide a meaningful description so that others may understand what the network is used for.

Fabulous network

SUBNETS

ID	NAME	RAIL	CIDR BLOCK	IP COUNT	STATUS	
588590d1	CIDR 1	Public	10.62.0.0/32	1	PreActive	
TOTAL IPv4s				1		

ADD A CIDR

NAME

RAIL

CIDR BLOCK

Enter CIDR name

Select rail

Enter CIDR block

Add row

Cancel

Save changes

EDITING A TSF-MSN NETWORK

You may change the network’s name or description, and add or remove network CIDRs. Once saved, changes to TSF-MSN networks will trigger updates to all associated access policies.

Downloading credentials for a TSF-MSN Network

The credentials for a TSF-MSN may be downloaded. Downloading application credentials for a given rail will generate new application credentials and revoke any previous application credentials for that rail.

To start, click the **download** icon.

NETWORKS

TSF-MSN NETWORKS

+ Create TSF-MSN network

ID	NAME	DESCRIPTION	LAST UPDATED	STATUS	
58847180	Network 1	Fabulous network	May 10, 2021 4:47 am	PreActive	

DOWNLOADING CREDENTIALS FOR A TSF-MSN NETWORK

A confirmation screen will be displayed.

ON THIS PAGE

- TSF Terms and Concepts

Managing Tenants

Creating a Tenant

Editing a Tenant’s Setup

Managing Networks

Choosing the network platform

Creating or importing an AWS VPC

Editing an AWS VPC

Deleting an AWS VPC

Creating a TSF-MSN Network

Activating a TSF-MSN Network

Editing a TSF-MSN Network

Downloading credentials for a TSF-MSN Network

Deleting a TSF-MSN Network

Creating a Comcast Legacy Network

Editing a Comcast Legacy Network

Deleting a Comcast

×

Confirm application credentials download

Download credentials for MF Aporeto 3's network Network 1

Network ID: f7eaa15e

Tenant's TSF Zone: DMZ

Rails:

Tenant's Data Classification: Public

Tip

Please see our [instructions](#) for installing TSF-MSN's application credentials.

Warning

Downloading application credentials for a given rail will generate new application credentials and revoke any previous application credentials for that rail.

RAIL

☒ Public☐ Protected☐ Private

Cancel

Download credentials

DOWNLOADING CREDENTIALS FOR A TSF-MSN NETWORK

Click the **Download credentials.** button to submit the request. Credentials will be downloaded to a file on your computer.

Deleting a TSF-MSN Network

A TSF-MSN network can be deleted if the following conditions are met:

- The network is in the pre-active or active state
- All TSF-MSN agents associated with the tenant are shut down or uninstalled
- The tenant has no services or access policies

To start, click the **delete** icon.

NETWORKS					
TSF-MSN NETWORKS					
+ Create TSF-MSN network					
ID	NAME	DESCRIPTION	LAST UPDATED	STATUS	
58847180	Network 1	Fabulous network	May 10, 2021 4:47 am	PreActive	

DELETING A TSF-MSN NETWORK

A confirmation screen will be displayed.

×

Confirm TSF-MSN network deletion

Are you sure you want to delete this network?

Network Name: Network 1

Tenant's TSF Zone: DMZ

Network ID: 58847180

Tenant's Data Classification: Public

Rails:

No, I changed my mind

Delete network

DELETING A TSF-MSN NETWORK

Click the **Delete network** button to submit the request. The network will enter the pending deletion state.

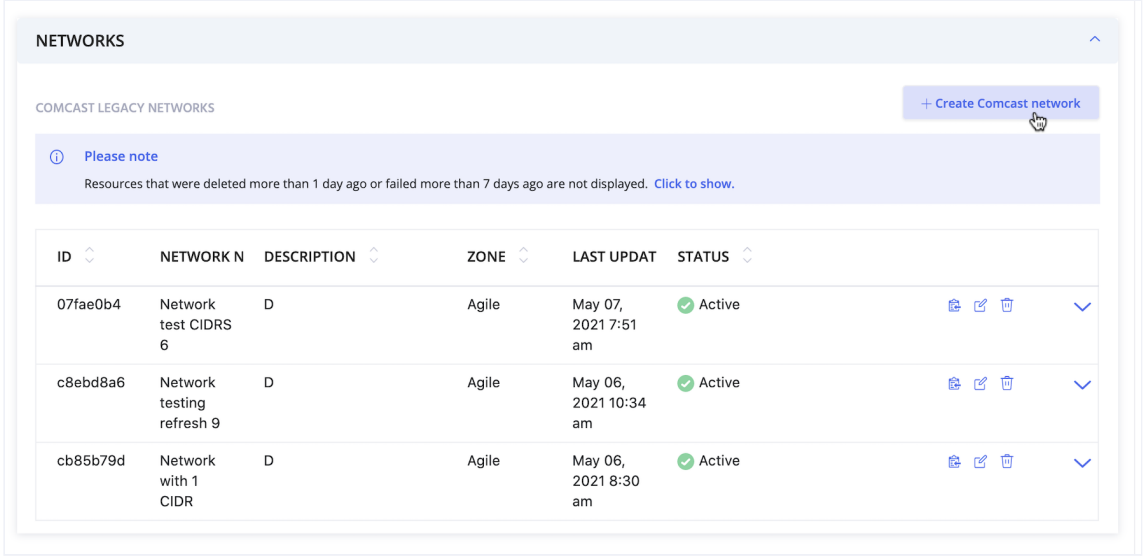
Creating a Comcast Legacy Network

- Note**
A Comcast legacy network in TSF is a mechanism to record the IP addresses or address blocks that host services on existing internal networks.

If you're creating the Comcast network in a new tenant, [follow the steps above](#) for choosing the platform, and pick Comcast legacy network.

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
- [Creating a Tenant](#)
- [Editing a Tenant's Setup](#)
- [Managing Networks](#)
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)



CREATING A COMCAST LEGACY NETWORK

The create Comcast legacy network form will be displayed.

Create Comcast legacy network

Add a Comcast legacy network to MF Comcast 4

Tenant's TSF Zone: DMZ

Tenant's Data Classification: Public

For more information ...

Instructions for creating a Comcast network in the OneCloud Portal can be found in [our user guide](#).

NETWORK NAME

The network name is shown to users in the OneCloud Portal. It should be concise and descriptive.

Network name

NETWORK IDENTIFIER

The identifier may not be changed. It is used to name resources by the TSF automation platforms.

Network ID

NETWORK DESCRIPTION

The description is shown to users in the OneCloud Portal. Please provide a meaningful description so that others may understand what the network is used for.

Description

CIDRS

NAME	RAIL	CIDR BLOCK	IP COUNT

ADD A CIDR

NAME

Enter CIDR name

RAIL

Select rail

CIDR BLOCK

Enter CIDR block

Add row

Cancel

Create network

CREATING A COMCAST LEGACY NETWORK

You will need to supply a network name, identifier, and description. See rules and guidance [here](#).

You will then need to add one or more CIDRs. A CIDR may represent a single host (i.e., /32) or an address block, and it must adhere to the same [rules that govern Comcast legacy networks](#).

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
- [Creating a Tenant](#)
- [Editing a Tenant's Setup](#)
- [Managing Networks](#)
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)

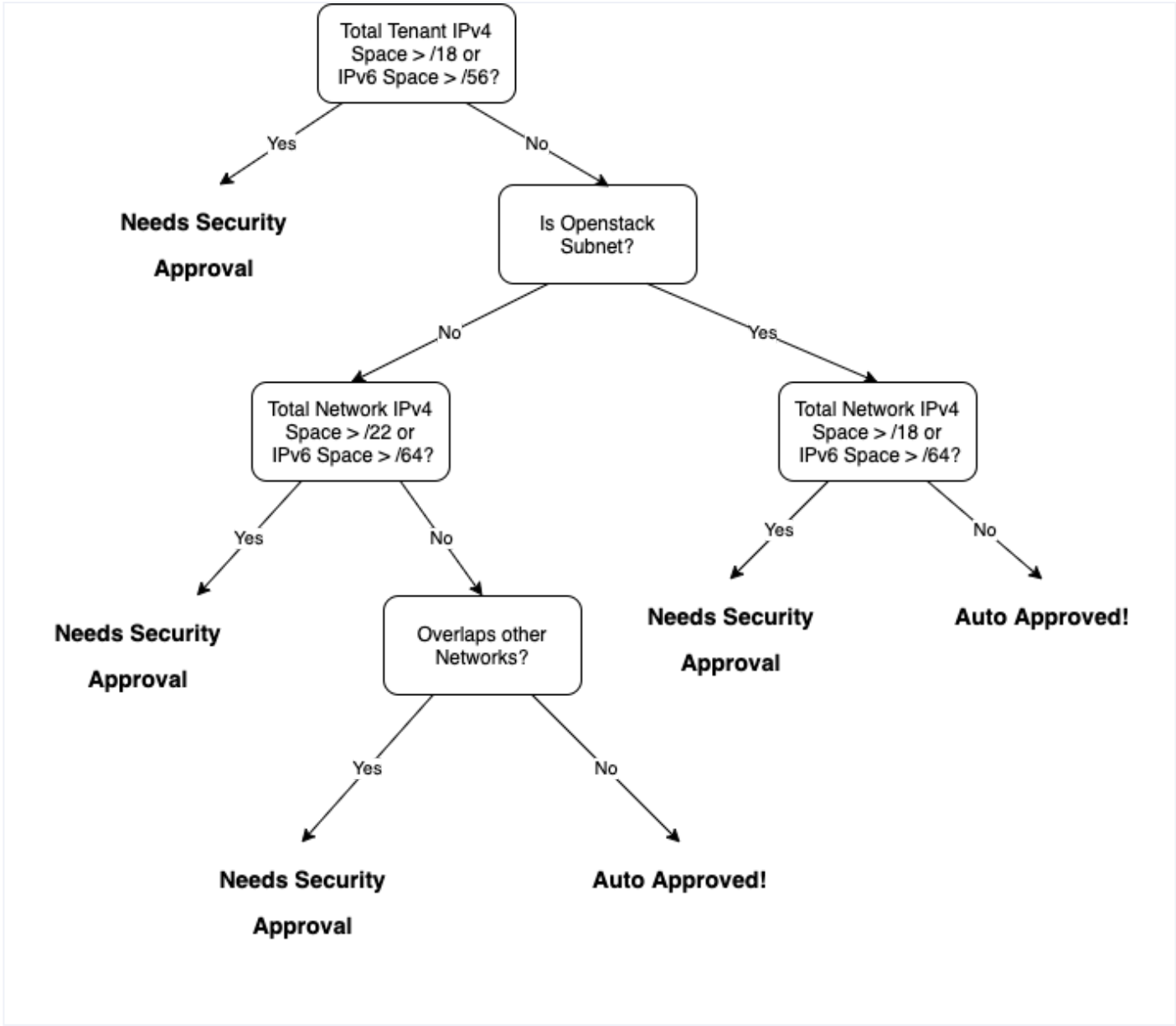
NETWORKS						
COMCAST LEGACY NETWORKS						
+ Create Comcast network						
NETWORK NAME	DESCRIPTION	ZONE	LAST UPDATED	STATUS	ID	
Network 1	D	Agile	Nov 16, 2021 3:35 pm	Pending Security Approval	feec0f0a	  

CREATING A COMCAST LEGACY NETWORK

Once the **Create network** button is hit, Comcast legacy networks go through a reviewal process before activation, as follows:

NETWORK STATUS	DEFINITION
Pending Security Approval	Network needs to be reviewed (rules specified in the graph below)
Active	Network immediately enters active state once auto-approved or reviewed and is available for inclusion in your tenant’s services

AUTO-APPROVAL AND SECURITY REVIEW REQUIRED CASES



COMCAST NETWORK APPROVAL FLOW

Editing a Comcast Legacy Network

Provided a TSF-MSN is in the active state, its name and description may be edited, and CIDRs may be added or removed.

Based on the modification, security review may be required as specified in [this graph](#).

To start, click the **edit** icon next to the network.

NETWORKS						
COMCAST LEGACY NETWORKS						
+ Create Comcast network						
NETWORK NAME	DESCRIPTION	ZONE	LAST UPDATED	STATUS	ID	
Network 1	D	Agile	Nov 16, 2021 3:29 pm	Active	feec0f0a	  

EDITING A COMCAST LEGACY NETWORK

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
 - [Creating a Tenant](#)
 - [Editing a Tenant’s Setup](#)
- [Managing Networks](#)
 - [Choosing the network platform](#)
 - [Creating or importing an AWS VPC](#)
 - [Editing an AWS VPC](#)
 - [Deleting an AWS VPC](#)
 - [Creating a TSF-MSN Network](#)
 - [Activating a TSF-MSN Network](#)
 - [Editing a TSF-MSN Network](#)
 - [Downloading credentials for a TSF-MSN Network](#)
 - [Deleting a TSF-MSN Network](#)
 - [Creating a Comcast Legacy Network](#)
 - [Editing a Comcast Legacy Network](#)
 - [Deleting a Comcast](#)

The edit form will be displayed.

×

Edit TSF-MSN network

Edit MF TSF-MSN tenant's network Network 1

Network ID: 58847180

Tenant's TSF Zone: DMZ

Rails: --

Tenant's Data Classification: Public

For more information ...

Instructions for updating TSF-MSN networks in the OneCloud Portal can be found in [our user guide](#).

NETWORK NAME

The network name is shown to users in the OneCloud Portal. It should be concise and descriptive.

Network 1

NETWORK IDENTIFIER

The identifier may not be changed. It is used to name resources by the TSF automation platforms.

network_1

NETWORK DESCRIPTION

The description is shown to users in the OneCloud Portal. Please provide a meaningful description so that others may understand what the network is used for.

Fabulous network

SUBNETS

ID	NAME	RAIL	CIDR BLOCK	IP COUNT	STATUS	
588590d1	CIDR 1	Public	10.62.0.0/32	1	PreActive	 
TOTAL IPv4s				1		
ADD A CIDR						
NAME		RAIL	CIDR BLOCK			
Enter CIDR name		Select rail	Enter CIDR block		Add row	

Cancel

Save changes

EDITING A COMCAST LEGACY NETWORK

Update the network’s name, description, and CIDRs as needed. Once saved, if the changes to the Comcast legacy network are approved, then the updates will cascade to the associated access policies.

Update the network’s name, description, and CIDRs as needed. Once you save your changes, they will be go through a reviewal process. If the changes are approved, the updates will cascade to the associated access policies.

Deleting a Comcast Legacy Network

A Comcast legacy network may be deleted provided it is in the active state and is not associated with any pending access policies.

To start, click the **delete** icon next to the desired network.

NETWORKS

COMCAST LEGACY NETWORKS

+ Create Comcast network

Please note

Resources that were deleted more than 1 day ago or failed more than 7 days ago are not displayed. [Click to show.](#)

ID	NETWORK	DESCRIPTION	ZONE	LAST UPDATE	STATUS	
07fae0b4	Network test CIDRS 6	D	Agile	May 07, 2021 7:51 am	Active	  
c8ebd8a6	Network testing refresh 9	D	Agile	May 06, 2021 10:34 am	Active	  
cb85b79d	Network with 1 CIDR	D	Agile	May 06, 2021 8:30 am	Active	  

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
- [Creating a Tenant](#)
- [Editing a Tenant’s Setup](#)
- [Managing Networks](#)
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)

A confirmation screen will be displayed. The services and access policies that would be affected by the deletion will be shown for review.

Confirm Comcast network deletion

Network ID: 5ba219c0

Tenant's TSF Zone: DMZ

Rails: Public

Tenant's Data Classification: Non-Public

AFFECTED SERVICES AND INBOUND ACCESS POLICIES

SERVICE	RAILS	PROTOCOL	PORTS	ICMP TYPE	CONSUMING TENANTS
DMZ Symantec SEP	Public	TCP	443		

AFFECTED OUTBOUND ACCESS POLICIES

PROVIDING TENANT	SERVICE	RAILS	PROTOCOL	PORTS	ICMP TYPE
MF AWS with IPv6 8	Service to accept then reject	Public	TCP	123	

No, I changed my mind

Delete network

DELETING COMCAST LEGACY NETWORK

Click the **Delete network** to start the deletion process. Once an Comcast legacy network is deleted, the associated access policies will be updated.

Viewing CIDR details

You may view the CIDR/segment details of any network, regardless of platform, by clicking on the large down-arrow next to right of the network.

NETWORKS

AWS VPCS

Create AWS VPCImport AWS VPC

Notice

Please note

ID	VPC NAME	VPC ID	DESCRIPTION	ACCOUNT	REGION	VPC BLOCKS	UPDATED	STATUS	
b9daeefc	network 2	vpc-23337cfa	Desc	Governance Test (26972674301)	us-east-1	96.102.64	May 05, 2021 12:02 pm	Active	

Subnet details for VPC network 2

SUBNET ID	SUBNET NAME	RAIL	AVAILABILITY ZONE	IPV4 CIDR BLOCK	IPV6 CIDR BLOCK	STATUS	
subnet-156e5d4f	public-0	Public	us-east-1a	96.102.64.16/28	2001:558:fd01::/64	Active	

49559c21Network testing refreshvpc-8c42e6e4DGovernance Test (26972674301)us-east-196.102.64May 02, 2021 9:53 amActive

VIEWING CIDR DETAILS

Managing Services

The **Services** pane displayed the services that your tenant has decided to use and/or share with other tenants.

Creating a Service

Before you can create a service, the tenant must have at least one active network. To create a service, click the **Create service** button in the services pane.

SERVICES

Create service

Please note

ID	SERVICE	DESCRIPTION	RAILS	PROTOCOL	PORTS/TYPE	LAST UPDATED	STATUS	
fd52e643	Protected service	D	Protected	ICMP	Echo	May 06, 2021 8:26 am	Active	

CREATING A SERVICE

ON THIS PAGE

- TSF Terms and Concepts**
- Managing Tenants**
- Creating a Tenant
- Editing a Tenant's Setup
- Managing Networks**
- Choosing the network platform
- Creating or importing an AWS VPC
- Editing an AWS VPC
- Deleting an AWS VPC
- Creating a TSF-MSN Network
- Activating a TSF-MSN Network
- Editing a TSF-MSN Network
- Downloading credentials for a TSF-MSN Network
- Deleting a TSF-MSN Network
- Creating a Comcast Legacy Network
- Editing a Comcast Legacy Network
- Deleting a Comcast

×

Create service

Add a service to MF Aporeto 2

ⓘ

For more information ...

Instructions for creating services in the OneCloud Portal can be found in [our user guide](#).

SERVICE DISPLAY NAME

The name is shown to users in the OneCloud Portal. It should be concise and descriptive.

SSH

SERVICE IDENTIFIER

The identifier is suggested for you, but edit below if needed. It may not be changed after service creation.

ssh

SERVICE DESCRIPTION

The description will be presented to users in the OneCloud portal. Please provide a meaningful description so that others may understand what the resource is.

This is an SSH service

RAILS

The rails describe where the service will be exposed. See [our user guide](#) for instructions on choosing a rail or rails for your service. Only those rails on active networks for this tenant are available for selection here.

☐ Public ☒ Protected ☐ Private

PROTOCOL

The protocol may not be changed after the service has been created.

ICMPv6

ICMP TYPE

The ICMP type may not be changed after the service has been created.

Echo

OWNER PRE-APPROVED

Provides owner pre-approval of tenant requests to access the service. Only a tenant owner can set this.

☐ Owner pre-approved

Cancel

Create service

CREATING A SERVICE

You will need to supply a service name, identifier, and description. See rules and guidance [here](#).

You will then select the rail(s) the service will be exposed on. Only those rails that are available on your networks are available for selection.

Next, you will select the service protocol. If the protocol is TCP or UDP, you will then be prompted to specify the ports. You can supply a single port (e.g., 443), a port range (e.g., 8000-9000), or a list of ports/port ranges (e.g., 80,443,8000-9000). If the protocol is ICMP or ICMPv6, you must specify the ICMP type. At this time, only Echo is available for selection.

If you want to pre-approve all access policy requests for your service, click the **Owner pre-approved** checkbox. If this checkbox is not checked, then any request to access the service must be explicitly approved or rejected by an owner of your tenant.

Once saved, the requested service definition will go to the NETO Security Change Review Group for review. Only after that team has reviewed and approved the service definition will it be transitioned to the active state and available for use in access policy definitions. Note that a service that is pending review by the Security Change Review Group may not be edited.

SERVICES							
ID	SERVICE	DESCRIPTION	RAILS	PROTOCOL	PORTS/TYPE	LAST UPDATED	STATUS
53ba7703	SSH	This is an SSH service	Protected	ICMPv6	Echo	May 10, 2021 4:18 am	Pending Security Approval Please contact the Security Change Review Group if not approved within two

ON THIS PAGE

[TSF Terms and Concepts](#)

[Managing Tenants](#)

[Creating a Tenant](#)

[Editing a Tenant's Setup](#)

[Managing Networks](#)

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

[Deleting a Comcast](#)

https://onecloud.comcast.net/docs/user-guide/management-console/tsf/#tsf-terms-and-concepts

25/36

You will see the status of your request change as it is approved or rejected.

SERVICES

+ Create service

Please note
Resources that were deleted more than 1 day ago or failed more than 7 days ago are not displayed. [Click to show.](#)

ID	SERVICE	DESCRIPTION	RAILS	PROTOCOL	PORTS/TYPE	LAST UPDATED	STATUS	
fd52e643	Protected service	D	Protected	ICMP	Echo	May 06, 2021 8:26 am	Active	  
a46a8ee3	SSH	Excellent service	Protected	TCP	8080	May 10, 2021 5:54 am	Security Rejected	   

Please contact the Security Change Review Group for assistance.

CREATING A SERVICE

Any comments the Security Change Review Group has made while approving or rejecting the service definition can be seen by clicking the **information** icon next to the service’s status.

View service history

View history for SSH

Service ID: a46a8ee3

Protocol: TCP

Ports/Type: 8080

SERVICE HISTORY

Records are displayed from newest to oldest.

UPDATED	BY	NAME	DESCRIPTIC	RAILS	STATUS	COMMENTS
May 10, 2021 5:54 am	mfranc213	SSH	Excellent service	Protected	Security Rejected	mfranc213: Not a good idea.

VIEWING SERVICE HISTORY

Editing a Service

A service in the active state may be edited by changing its description or by adding or removing rails. You can also change the **Owner Pre-Approved** setting of the service.

To start, click the **edit** icon next to the service.

SERVICES

+ Create service

ID	SERVICE	DESCRIPTIO	RAILS	PROTOCOL	PORTS/TYPI	LAST UPDATED	STATUS	
53ba7703	SSH	This is an SSH service	Protected	ICMPv6	Echo	May 10, 2021 4:24 am	Active	  

EDITING A SERVICE

The edit service form will be displayed.

ON THIS PAGE

- TSF Terms and Concepts
- Managing Tenants
 - Creating a Tenant
 - Editing a Tenant’s Setup
- Managing Networks
 - Choosing the network platform
 - Creating or importing an AWS VPC
 - Editing an AWS VPC
 - Deleting an AWS VPC
 - Creating a TSF-MSN Network
 - Activating a TSF-MSN Network
 - Editing a TSF-MSN Network
 - Downloading credentials for a TSF-MSN Network
 - Deleting a TSF-MSN Network
 - Creating a Comcast Legacy Network
 - Editing a Comcast Legacy Network
 - Deleting a Comcast

×

Edit service

Edit MF Aporeto 2's SSH service

Service ID: 53ba7703

ⓘ

For more information ...

Instructions for updating services in the OneCloud Portal can be found in [our user guide](#).

SERVICE DISPLAY NAME

The service display name can be updated to reflect changes after the network has been created.

SSH

SERVICE IDENTIFIER

The identifier may not be changed. It is used to name resources by the TSF automation platforms.

ssh

SERVICE DESCRIPTION

The description will be presented to users in the OneCloud portal. Please provide a meaningful description so that others my understand what the resource is.

This is an SSH service

RAILS

The rails describe where the service will be exposed. See [our user guide](#) for instructions on choosing a rail or rails for your service. Only those rails on active networks for this tenant are available for selection here.

☐ Public ☒ Protected ☐ Private

PROTOCOL

The protocol may not be changed after the service has been created.

ICMPv6

ICMP TYPE

The ICMP type may not be changed after the service has been created.

Echo

OWNER PRE-APPROVED

Provides owner pre-approval of tenant requests to access the service. Only a tenant owner can set this.

☐ Owner pre-approved

SECURITY PRE-APPROVED

Provides security pre-approval of tenant requests to access the service. Only a security reviewer can set this.

☐ Security pre-approved

Cancel

Save changes

EDITING A SERVICE

Similar to service definition creation, service modifications are reviewed by the NETO Security Change Review Group. Once a service definition modification has been approved, any needed updates to associated access policies will occur automatically.

Proposed changes as well as any comments the Security Change Review Group has made while approving or rejecting the service definition modification can be seen by clicking the **information** icon next to the status.

SERVICES

ⓘ Please note

Resources that were deleted more than 1 day ago or failed more than 7 days ago are not displayed. [Click to show.](#)

ID	SERVICE	DESCRIPTION	RAILS	PROTOCOL	PORTS/TYPE	LAST UPDATED	STATUS	
fd52e643	Protected service	D	Protected	ICMP	Echo	May 06, 2021 8:26 am	Active / Pending Security Approval	
a46a8ee3	SSH	Excellent service	Protected	TCP	8080	May 10, 2021 5:54 am	Security Rejected Please contact the Security Change Review Group for assistance.	

EDITING A SERVICE

The service history dialogue will be displayed.

ON THIS PAGE

- [TSF Terms and Concepts](#)**
- [Managing Tenants](#)**
- [Creating a Tenant](#)
- [Editing a Tenant's Setup](#)
- [Managing Networks](#)**
- [Choosing the network platform](#)
- [Creating or importing an AWS VPC](#)
- [Editing an AWS VPC](#)
- [Deleting an AWS VPC](#)
- [Creating a TSF-MSN Network](#)
- [Activating a TSF-MSN Network](#)
- [Editing a TSF-MSN Network](#)
- [Downloading credentials for a TSF-MSN Network](#)
- [Deleting a TSF-MSN Network](#)
- [Creating a Comcast Legacy Network](#)
- [Editing a Comcast Legacy Network](#)
- [Deleting a Comcast](#)

View service history

View history for Protected service

Service ID: fd52e643

Protocol: ICMP

Ports/Type: Echo

SERVICE HISTORY

Records are displayed from newest to oldest.

UPDATED	BY	NAME	DESCRIPTIC	RAILS	STATUS	COMMENTS
May 06, 2021 8:26 am	system	Protected service	D	Protected	Active	
May 10, 2021 6:01 am	mfranc213	Protected service	Better description	Protected, Private	Pending Security Approval	

EDITING A SERVICE

Deleting a service

Provided it has no access policies pending for it, a service may be deleted.

To start, click the **delete** icon next to the desired service.

SERVICES

+ Create service

ID	SERVICE	DESCRIPTIC	RAILS	PROTOCOL	PORTS/TYPI	LAST UPDATED	STATUS	
53ba7703	SSH	This is an SSH service	Protected	ICMPv6	Echo	May 10, 2021 4:24 am	Active	

DELETING A SERVICE

A confirmation screen will be displayed.

Confirm service deletion

Are you sure you want to delete MF Aporeto 2's SSH service?

Service ID: 53ba7703

No, I changed my mind

Delete service

DELETING A SERVICE

Click the **Delete service** to start the deletion process. Once an AWS VPC is marked for deletion, any associated access policies will be removed.

Managing Access Policies

Access policies are the engine of the TSF system, the mechanism by which service access is provided by one tenant to another. An inbound access policy for a providing tenant corresponds to an outbound access policy for the consuming tenant.

Creating an Inbound Access Policy

The **Inbound access policies** pane shows the services in your tenant that have been, or are in the process of being, exposed to other tenants. Access policies may be created by either the providing or the consuming tenant. To create an access policy as a providing tenant, click the **Create inbound access policy** button.

INBOUND ACCESS POLICIES

+ Create inbound access policy

This tenant has no inbound access policies.

ON THIS PAGE

[TSF Terms and Concepts](#)

[Managing Tenants](#)

[Creating a Tenant](#)

[Editing a Tenant's Setup](#)

[Managing Networks](#)

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

[Deleting a Comcast](#)

The inbound access policy form will be displayed.

Create inbound access policy

Provide access to this tenant's MF Comcast 4's service



For more information ...

Instructions for creating an inbound access policy in the OneCloud Portal can be found in [our user guide](#).

CONSUMING TENANT

Only tenants with at least one active network are shown.

MY SERVICES

Only active services are shown.

Cancel

Provide access

CREATING AN INBOUND ACCESS POLICY

Select the consuming tenant, and the service from your tenant that will be exposed, and click the **Provide access** button. The access policy request will then enter the pending security approval state and will be submitted to the NETO Security Change Review Group for review. See the [Access Policy Approval and Activation](#) section below for more information on the approval process. You will see the status of the access policy change as it is approved or rejected.

Once an access policy request is fully approved, changes will be provisioned by TSF automation. When the changes are successfully provisioned, the access policy will be marked active and the requesting tenant will be able to make use of your service.

Approving or Rejecting an Inbound Access Policy Request [🔗](#)

If an inbound access policy request has been made against a service that was not marked pre-approved by the owner, the request will appear with **approve** and **reject** icons.

INBOUND ACCESS POLICIES						
+ Create inbound access policy						
ACCESS POLICY ID	MY SERVICE	CONSUMING TENANT	LAST UPDATED	STATUS		
6c5ab888	Protected service	MF AWS no IPv6 2	May 10, 2021 6:06 am	Pending Owner Approval Please contact the service owner if not approved within three business days.		

REVIEWING INBOUND ACCESS POLICY REQUEST

To approve the request, click the **approve** icon. The approval form will be displayed.

Confirm access policy approval

Are you sure you want to grant access for the tenant MF AWS no IPv6 2 to the Protected service service?

Access Policy ID: 6c5ab888 📄

ⓘ Please note

A request for inbound access will be reviewed by the security team once approval has been given by a tenant owner.

REASON

Enter the reason you are approving access.

No, I changed my mindApprove access policy

APPROVING AN INBOUND ACCESS POLICY REQUEST

Optionally supply a comment, and hit the **Approve access** button.

To deny the request, click the **reject** icon. A confirmation form will be displayed.

ON THIS PAGE

TSF Terms and Concepts

Managing Tenants

Creating a Tenant

Editing a Tenant's Setup

Managing Networks

Choosing the network platform

Creating or importing an AWS VPC

Editing an AWS VPC

Deleting an AWS VPC

Creating a TSF-MSN Network

Activating a TSF-MSN Network

Editing a TSF-MSN Network

Downloading credentials for a TSF-MSN Network

Deleting a TSF-MSN Network

Creating a Comcast Legacy Network

Editing a Comcast Legacy Network

Deleting a Comcast

×

Confirm access policy rejection

Are you sure you want to reject MF AWS no IPv6 2's access to the Protected service service?

Access Policy ID: 6c5ab888

REASON

Enter the reason you are rejecting access.

No, I changed my mind

Reject access policy

REJECTING AN INBOUND ACCESS POLICY REQUEST

Optionally supply a comment, and hit the **Reject access** button.

Viewing consuming tenant details

You may view the consuming tenant details of an inbound access policy by clicking on the large down-arrow next to right of the policy.

INBOUND ACCESS

+ Create inbound access request

Please note

Resources that were deleted more than 1 day ago or failed more than 7 days ago are not displayed. [Click to show.](#)

MY SERVICE	CONSUMING TENANT	LAST UPDATED	STATUS	ACCESS POLICY ID			
Protected Service	Filtering Demo Non Owner	Sep 16, 2021 12:34 pm	Active	cff0d218			
Consuming tenant details							
PLATFORM	ZONE	DATA CLASSIFICATION	IP ADDRESS SPACE	RAILS	OWNERS	ID	
Comcast legacy	DMZ	Public	100.88.0.0/31, 100.88.0.2/32	Public, Protected, Private	Couja, Salem, Howard, V	9f1f1012e	
Public Service	Filtering Demo Non Owner	Jul 22, 2021 11:21 pm	Active	661e635c			

VIEWING CONSUMING TENANT DETAILS FOR AN INBOUND ACCESS POLICY

Deleting an inbound access policy

Provided it is not in a pending state, an inbound access policy may be deleted.

To start, click the **delete** icon next to the desired access policy.

INBOUND ACCESS POLICIES

+ Create inbound access policy

ACCESS POLICY ID	MY SERVICE	CONSUMING TENANT	LAST UPDATED	STATUS	
6c5ab888	Protected service	MF AWS no IPv6 2	May 10, 2021 6:08 am	Active	

DELETING AN INBOUND ACCESS POLICY

A confirmation screen will be displayed.

×

Confirm inbound access policy deletion

Are you sure you want to remove MF AWS no IPv6 2's access to this tenant's Protected service service?

Access Policy ID: 6c5ab888

No, I changed my mind

Delete access policy

DELETING AN INBOUND ACCESS POLICY

Click the **Delete** to start the deletion process.

Creating an Outbound Access Policy

The **Outbound access policies** pane shows the services from other tenants that have been, or are in the process of being, exposed to your tenant. If you want to request access to another tenant’s service, click the **Create outbound access policy** button.

ON THIS PAGE

TSF Terms and Concepts

Managing Tenants

[Creating a Tenant](#)

[Editing a Tenant’s Setup](#)

Managing Networks

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

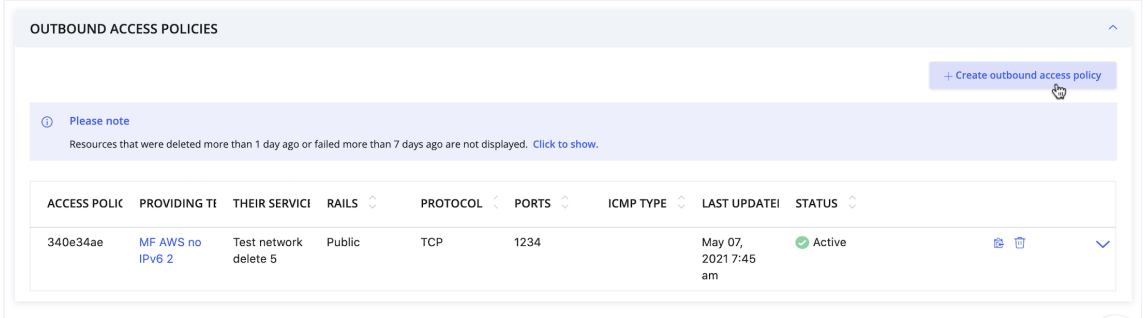
[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

[Deleting a Comcast](#)



CREATING AN OUTBOUND ACCESS POLICY REQUEST

The create outbound access policy form will be displayed.

Create outbound access policy

Request MF Comcast 4's access to a service

For more information ...

Instructions for creating an outbound access policy in the OneCloud Portal can be found in [our user guide](#).

PROVIDING TENANT

A providing tenant must have at least one active service.

MF AWS with IPv6 8

SERVICE

Only active services of the selected tenant are shown.

Service 2

Cancel

Request access

CREATING AN OUTBOUND ACCESS POLICY REQUEST

Select the providing tenant and service and click the **Request access** button.

Once your request is created, the request will go through the approval process. See the [Access Policy Approval and Activation](#) section below for details.

If an outbound access policy has been waiting for service owner approval for more than seven days, please contact the service owners. An email link is provided for you so you can do this easily.

OUTBOUND ACCESS								
+ Create outbound access request								
PROVIDING TENANT	THEIR SERVICE	RAILS	PROTOCOL	PORTS	ICMP TYPE	LAST UPDATED	STATUS	ACCESS POLICY ID
Atlas	Atlas	Public	TCP	80		Oct 12, 2021 7:07 am	Pending Owner Approval	5c3d5fa2
This request is only 3 days old. If it has not been approved by October 16, 2021, please contact the service owners.								
Internal DNS	Internal TCP DNS	Public	TCP	53		Oct 12, 2021 7:08 am	Pending Owner Approval	5c3dc2da
This request is already 10 days old. Please email the service owners .								
Internal DNS	Internal UDP DNS	Public	UDP	53		Mar 26, 2021 4:16 pm	Active	5c3e1f46

CONTACTING SERVICE OWNERS

When the access policy is fully approved changes will automatically be provisioned by TSF automation. Once those changes are successfully provisioned the access policy will be listed as active and your tenant will be able to make use of that service.

Viewing providing tenant details

You may view the providing tenant details of an outbound access policy by clicking on the large down-arrow next to right of the policy.

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
 - [Creating a Tenant](#)
 - [Editing a Tenant's Setup](#)
- [Managing Networks](#)
 - [Choosing the network platform](#)
 - [Creating or importing an AWS VPC](#)
 - [Editing an AWS VPC](#)
 - [Deleting an AWS VPC](#)
 - [Creating a TSF-MSN Network](#)
 - [Activating a TSF-MSN Network](#)
 - [Editing a TSF-MSN Network](#)
 - [Downloading credentials for a TSF-MSN Network](#)
 - [Deleting a TSF-MSN Network](#)
 - [Creating a Comcast Legacy Network](#)
 - [Editing a Comcast Legacy Network](#)
 - [Deleting a Comcast](#)

https://onecloud.comcast.net/docs/user-guide/management-console/tsf/#tsf-terms-and-concepts

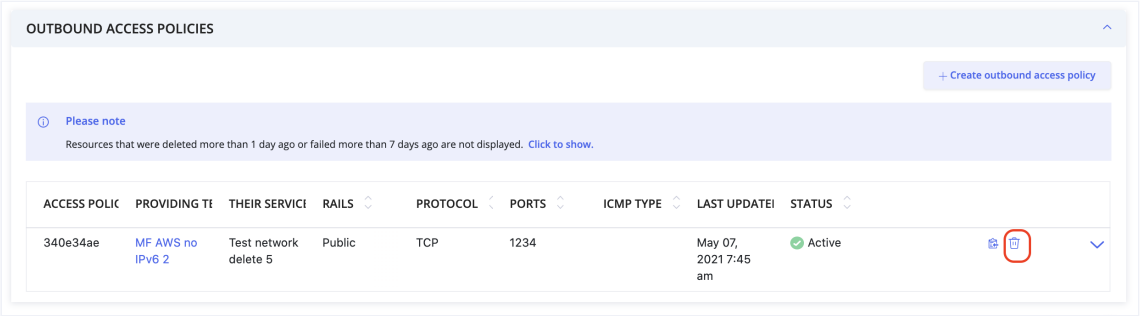
31/36

VIEWING PROVIDING TENANT DETAILS FOR AN OUTBOUND ACCESS POLICY

Deleting an outbound access policy

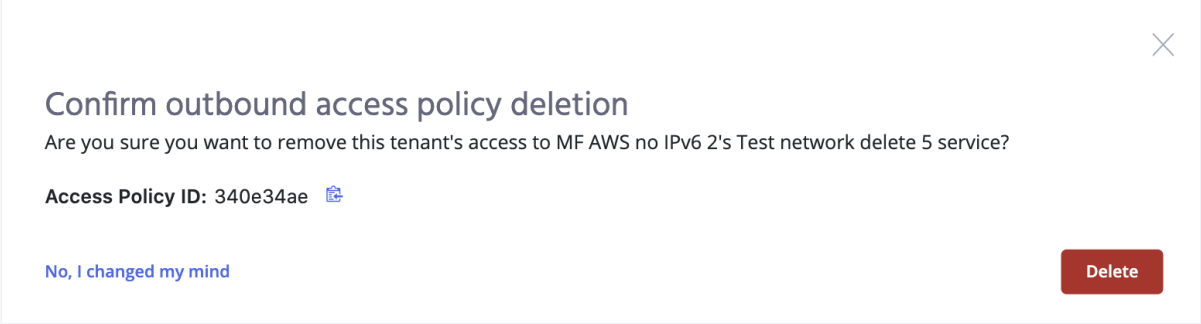
Provided it is not in a pending state, an outbound access policy may be deleted.

To start, click the **delete** icon next to the desired access policy.



DELETING AN OUTBOUND ACCESS POLICY

A confirmation screen will be displayed.



DELETING AN OUTBOUND ACCESS POLICY

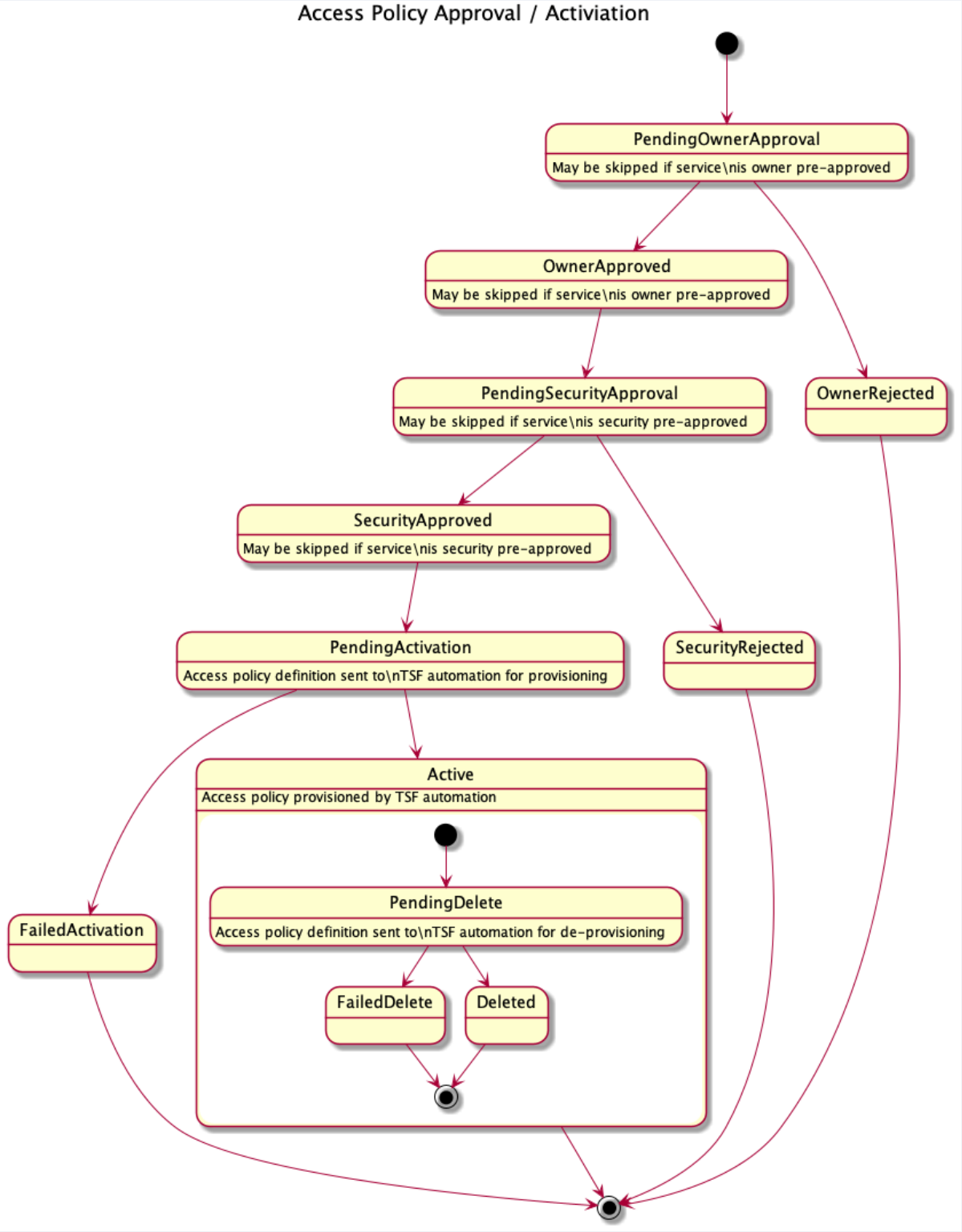
Click the **Delete** to start the deletion process.

Access Policy Approval and Activation

Before being provisioned by TSF automation, an access policy may require approval from both the owner of the providing service’s tenant and from the security team. The follow state diagram shows this approval and activation flow.

ON THIS PAGE

- TSF Terms and Concepts**
- Managing Tenants**
- Creating a Tenant
- Editing a Tenant’s Setup
- Managing Networks**
- Choosing the network platform
- Creating or importing an AWS VPC
- Editing an AWS VPC
- Deleting an AWS VPC
- Creating a TSF-MSN Network
- Activating a TSF-MSN Network
- Editing a TSF-MSN Network
- Downloading credentials for a TSF-MSN Network
- Deleting a TSF-MSN Network
- Creating a Comcast Legacy Network
- Editing a Comcast Legacy Network
- Deleting a Comcast



ON THIS PAGE

TSF Terms and Concepts

Managing Tenants

[Creating a Tenant](#)

[Editing a Tenant's Setup](#)

Managing Networks

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

[Deleting a Comcast](#)

The TSF registry has many [pre-allowed and pre-approved](#) tenants and services. Your tenant may request access to these services or provide access to these tenants, without having to go through the normal approval process.

SERVICES								
ID	SERVICE	DESCRIPTION	RAILS	PROTOCOL	PORTS/TYPE	LAST UPDATED	STATUS	
a739d0c4	Test network delete 5	Desc	Public	TCP	1234	May 01, 2021 4:51 am	Active	The tenant owner has pre-approved access policy requests for this service. The security team has pre-approved access policy requests for this service.

PRE-APPROVED ACCESS POLICIES FOR A SERVICE

Resources Managed By TSF Automation to Provide Access to Services

TSF automation currently manages the following [policy enforcement points](#) (PEPs):

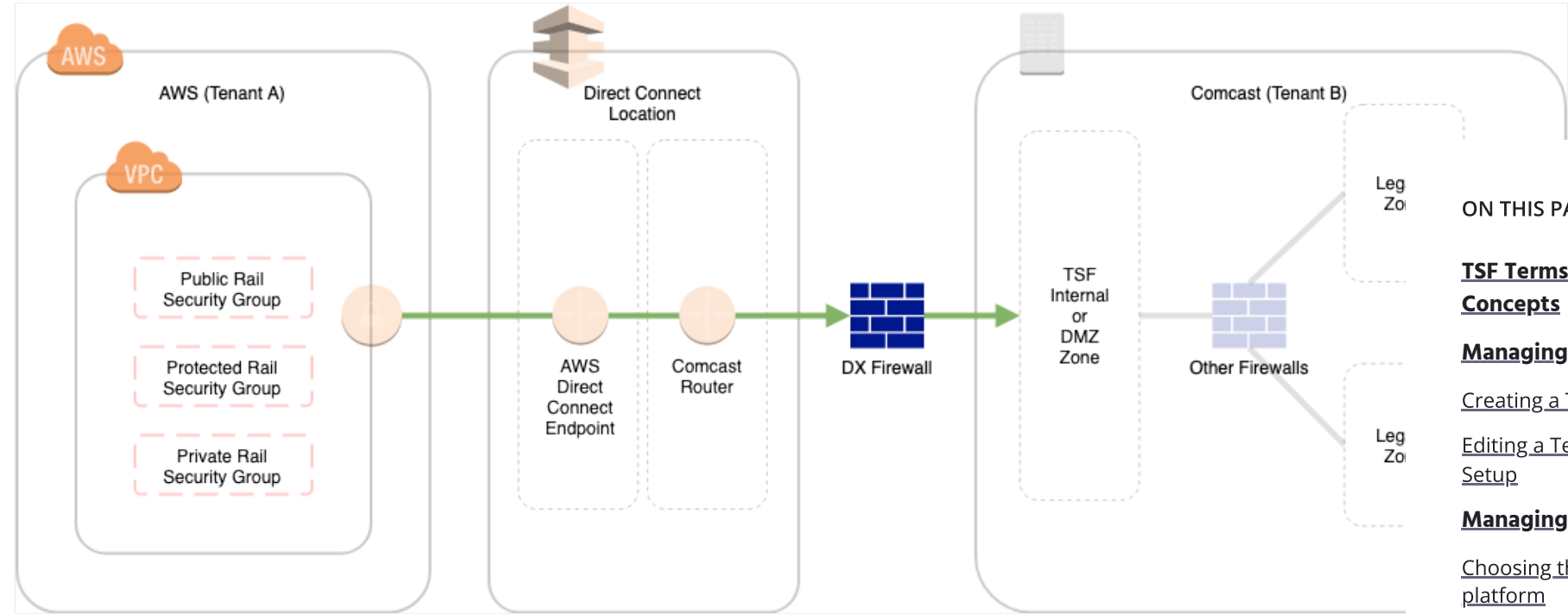
- AWS security groups
- Direct connect (DX) firewalls between AWS and Comcast legacy networks

Which of these resources needs to be changed in order to provide service access depends on the networks involved in the access policy request, as described next:

AWS TO COMCAST

Scenario 1: *AWS tenant* accessing service on a *Comcast legacy network*

TSF automation allows an AWS tenant (Tenant A) to access a service provided by a

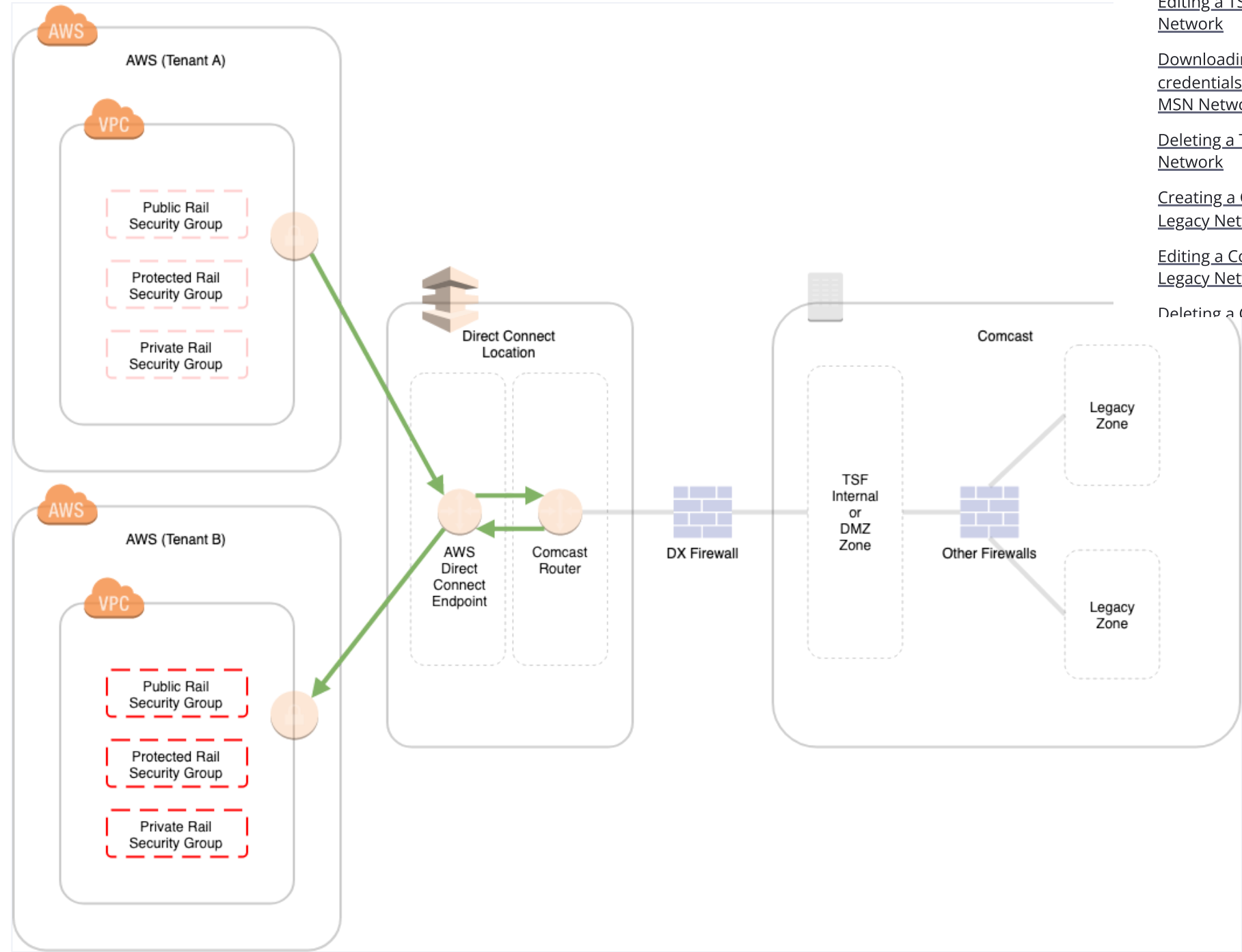


TSF PEPS - AWS TO COMCAST

AWS TO AWS

Scenario 2: AWS tenant accessing service on an AWS VPC

TSF automation allows an AWS tenant (Tenant A) to access a service provided by another AWS tenant (Tenant B) by modifying the AWS security groups for Tenant B:



TSF PEPS - AWS TO AWS

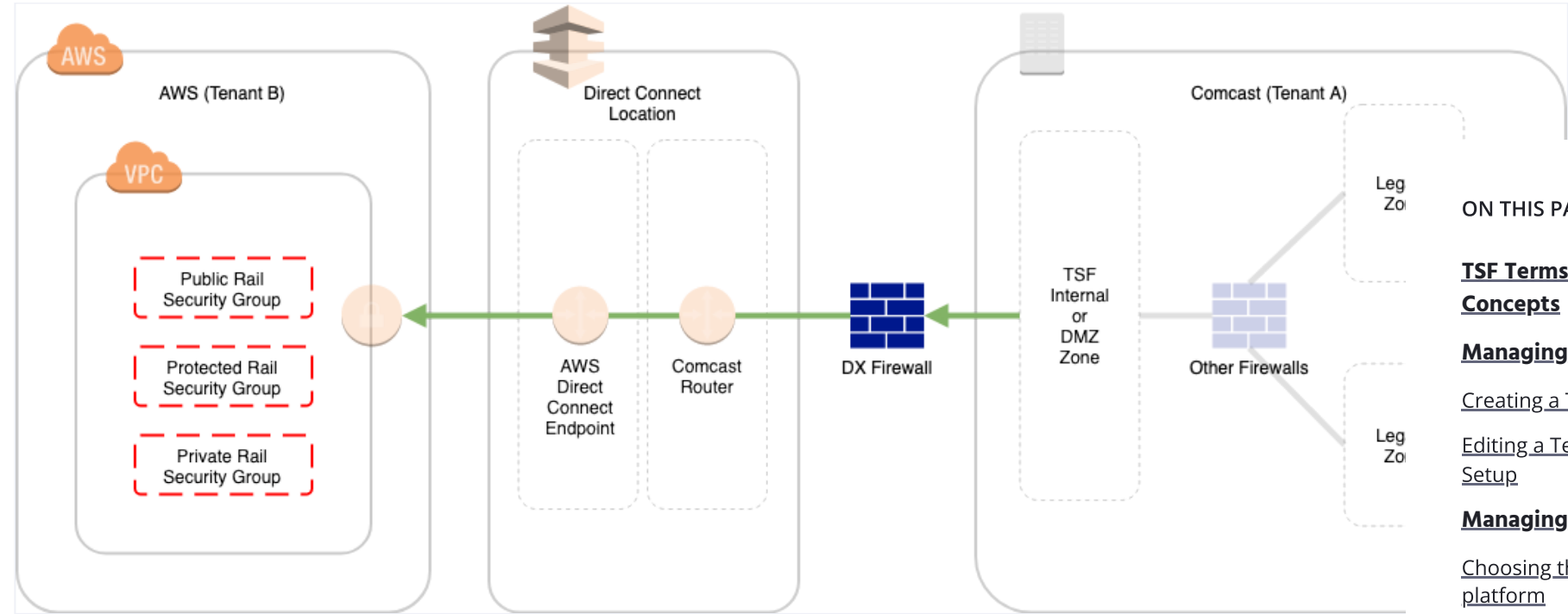
COMCAST TO AWS

Scenario 3: Comcast tenant accessing service on an AWS VPC

TSF automation allows a tenant on a Comcast legacy network (Tenant A) to access a service provided by an AWS tenant (Tenant B) by modifying the DX firewall and AWS security groups for Tenant B:

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
 - [Creating a Tenant](#)
 - [Editing a Tenant's Setup](#)
- [Managing Networks](#)
 - [Choosing the network platform](#)
 - [Creating or importing an AWS VPC](#)
 - [Editing an AWS VPC](#)
 - [Deleting an AWS VPC](#)
 - [Creating a TSF-MSN Network](#)
 - [Activating a TSF-MSN Network](#)
 - [Editing a TSF-MSN Network](#)
 - [Downloading credentials for a TSF-MSN Network](#)
 - [Deleting a TSF-MSN Network](#)
 - [Creating a Comcast Legacy Network](#)
 - [Editing a Comcast Legacy Network](#)
 - [Deleting a Comcast](#)

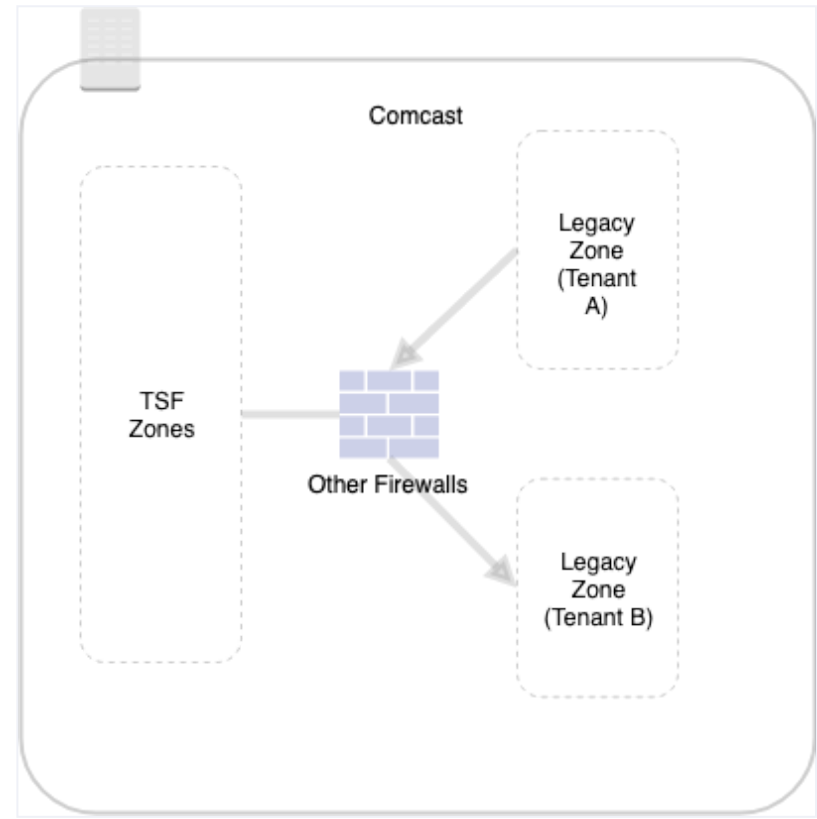


TSF PEPS - COMCAST TO AWS

COMCAST TO COMCAST

Scenario 4: Comcast tenant accessing service on a Comcast legacy network

TSF automation currently *can not make changes to firewalls or other PEPs* to allow a tenant on a Comcast legacy network (Tenant A) to access a service provided by another tenant on a Comcast legacy network (Tenant B). A traditional firewall request will need to be completed if firewall modifications are required for connectivity under this scenario.



TSF PEPS - COMCAST TO COMCAST

TSF Resource Names, Identifiers, and Descriptions

As you create and edit tenants, networks, services and access policies in OneCloud, you will see common fields that need to be supplied. These fields are governed by the same requirements and guidelines, as described here.

You will need to supply a name for the resource. The name is shown to users in the OneCloud Portal and should be concise and descriptive. The name may contain alphanumeric characters, underscores, or spaces ([0-9A-Za-z_]). The name may NOT be updated after the resource has been created.

A resource identifier will be generated from the supplied name. The identifier will be used throughout the backend systems and, while you may change it during the creation of the resource, you may not change it after the request to create the resource has been submitted.

ON THIS PAGE

- [TSF Terms and Concepts](#)
- [Managing Tenants](#)
 - [Creating a Tenant](#)
 - [Editing a Tenant's Setup](#)
- [Managing Networks](#)
 - [Choosing the network platform](#)
 - [Creating or importing an AWS VPC](#)
 - [Editing an AWS VPC](#)
 - [Deleting an AWS VPC](#)
 - [Creating a TSF-MSN Network](#)
 - [Activating a TSF-MSN Network](#)
 - [Editing a TSF-MSN Network](#)
 - [Downloading credentials for a TSF-MSN Network](#)
 - [Deleting a TSF-MSN Network](#)
 - [Creating a Comcast Legacy Network](#)
 - [Editing a Comcast Legacy Network](#)
 - [Deleting a Comcast](#)

Finally, you will need to supply a description of the resource. The description is shown to users in the OneCloud Portal. Provide a meaningful description so that others may understand what the resource is used for.

Additional Resources

[AWS VPC Documentation](#)

[Security Groups](#)

[TSF Documentation](#)

[Comcast legacy network segmentation](#) eLearning course on Workday Learning

ON THIS PAGE

TSF Terms and Concepts

Managing Tenants

[Creating a Tenant](#)

[Editing a Tenant’s Setup](#)

Managing Networks

[Choosing the network platform](#)

[Creating or importing an AWS VPC](#)

[Editing an AWS VPC](#)

[Deleting an AWS VPC](#)

[Creating a TSF-MSN Network](#)

[Activating a TSF-MSN Network](#)

[Editing a TSF-MSN Network](#)

[Downloading credentials for a TSF-MSN Network](#)

[Deleting a TSF-MSN Network](#)

[Creating a Comcast Legacy Network](#)

[Editing a Comcast Legacy Network](#)

[Deleting a Comcast](#)